# Reading Tea Leaves: Benefits and Costs of Learning about Future Demand Shocks*

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**Abstract**

In the noisy rational expectations equilibrium (NREE) literature, existing studies have investigated acquisition of information about current demand shocks and current and future dividend innovations. This article fills an important gap by characterizing the equilibrium effects of learning about future demand shocks. It shows that with access to additional information about future demand shocks, investors learn less about current dividend innovations and demand shocks and allocate greater precision to signals about future demand shocks through the front-running channel and a novel *future uncertainty channel*. This future uncertainty channel operates on two levels and breaks the fallacy of composition: at the individual level, acquisition of information about future demand shocks reduces the conditional variance of future excess payoffs; in the aggregate, market-wide dividend information acquisition declines, rendering dividends riskier and raising risk premiums. This article also demonstrates that greater heterogeneity in private information increases price volatility and that learning about future demand shocks renders prices less informative contemporaneously but more informative in later periods.

*Keywords:* NREE, future demand shocks, information acquisition, fallacy of composition

*JEL:* D82, D83, G11, G14

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## 1. Introduction

Noisy rational expectations equilibrium (NREE) models have served as an established framework for information choice problems in the study of financial markets. These models usually introduce exogenous noise trades, alternatively termed shocks to the demand for or supply of risky assets, to prevent their prices from fully revealing private information observed by investors (Grossman and Stiglitz, 1980; Hellwig, 1980; Verrecchia, 1982). These shocks are orthogonal to dividend innovations and do not affect the fundamental value, or liquidation value, of the risky assets. It is now well understood that noise trades can be micro-founded by hedging needs (Liu and Wang, 2016; Medrano and Vives, 2004), liquidity needs (Han et al., 2016), or sentiment (Shleifer and Summers, 1990). This article abstracts from micro-foundations of noise trades and hereinafter refers to them as demand shocks.

Intuitively, learning about future demand shocks affects the financial markets via the front-running channel: if investors can learn about future shocks, they have incentives to buy (sell) shares now and sell (buy) them tomorrow if tomorrow's demand is predicted to be high (low). In this article, I propose a *future uncertainty channel*, through which future demand shocks affect the market regardless of whether the front-running motive is present. If we shut down the front-running channel (e.g., in a case of a stream of dividend strips), at first glance, future demand shocks that cannot be learned about today, *should* be irrelevant to investor behavior and the current asset price. The future uncertainty channel, in fact, makes the future relevant to the present when there is acquisition of information about future demand shocks. The future uncertainty channel works through two forces to increase the investor's ex ante utility: at the aggregate level, market-wide dividend information acquisition decreases, which makes dividends riskier with higher returns; at the individual level, acquisition of information about future demand shocks makes the conditional variance of future excess payoffs lower. Higher expected returns and lower conditional variance of future excess payoffs both increase investors' ex ante utility.

In the literature, investors are usually worse off if everyone holds more private information. This is a manifestation of *fallacy of composition*, i.e. the error of assuming what is true of a member of a group is true for the group as a whole. In the context of financial markets, more private information is better for an atomic individual investor. In the aggregate, however, when all investors acquire more private information, assets become less risky with lower returns, as discussed by Kurlat and Veldkamp (2015). Learning about current demand shocks is not sufficient to break the fallacy; but learning about *future* demand shocks does. It creates heterogeneity between generations, introduces risk sharing across time and information spillovers between generations, and reduces market-wide dividend information acquisition, thus making the asset riskier with a higher risk premium.

Acquiring a signal about future demand shocks also echoes the general idea of news-driven business cycles in the macroeconomic literature. News shocks mainly concern future total factor productivity, ignoring an important aspect of price formation: demand shocks can drive the prices of goods away from their fundamental values. Learning about future demand shocks is even more important when we turn our attention to the financial market, which is the focus of this article.

This article studies how learning about future demand shocks affects investor behavior, including choices of information and investment, and market outcomes, including price informativeness and volatility. Specifically, I propose a model of a competitive CARA-normal economy with overlapping generations of investors who live for two periods. In each period, investors first decide, subject to a linear information capacity constraint, how to allocate precision to signals on current dividend innovations, current demand shocks and future demand shocks. Then they make subsequent investment decisions after observing signal realizations and the equilibrium asset price. To disentangle the front-running channel and the future uncertainty channel, I study two versions of the model, one with a long-lived asset ("dynamic model") and one with dividend strips ("static model").

The model features risk sharing across time and information spillover between generations.

On the one hand, with signals of future demand shocks, future risks are incorporated into today's price. As a result, tomorrow's market risks are partially shared by today's investors. On the other hand, only the young generation optimally chooses to learn about future demand shocks. They will be able to extract *more* information about dividend innovations from the equilibrium price in their second period, but *less* in the current period. The old generation knows more about the current demand shocks from the previous period, and their private information is aggregated by and then spills over to the young generation through the equilibrium price.

This model also features two margins of information choices. On the intensive margin, learning about tomorrow's demand shocks today and learning about them tomorrow are imperfect substitutes, with the former also affecting the marginal value of learning tomorrow. Assuming learning about future demand shocks is equally costly today and tomorrow, learning about tomorrow's demand shock today can fully substitute for learning about it tomorrow and offer an extra benefit from a higher marginal value of learning. On the extensive margin, we are weighing the cost-adjusted marginal benefits of one unit of precision allocated to signals about dividend innovations and demand shocks. The extra benefit provided by learning about future demand shocks makes learning about them more beneficial than learning about current dividend innovations. Therefore, with information acquisition equally costly (both across signals and for each unit of precision of the same signal), investors specialize in information acquisition and optimally choose to solely acquire signals about future demand shocks in the first period and signals about dividend innovations in the second period. On the aggregate level, less dividend information is learned and thus price informativeness decreases, while price volatility increases due to more heterogeneous private information.

To put the question into an empirical context, I discuss a few examples where learning about future demand shocks plays an important role. One notable example is index recomposition. As Han et al. (2016) points out, institutional traders, e.g. index

funds or ETFs, need to rebalance portfolios for non-informational reasons around index recomposition. The existence of index funds and the requirement that they replicate indices create demand shocks that are non-fundamental to the underlying securities. On the announcement date of index recomposition, investors receive a signal of effective-date demand shocks. Sophisticated investors can even adjust portfolios before the index recomposition announcement by making informed predictions about how index funds will have to rebalance. Another set of examples are in a similar vein as the news shock, except that here the news that investors receive is not about future total factor productivity, but about future demand shocks. One case with far-reaching impact is the BlackRock-BGI acquisition. BlackRock, Inc. announced on June 11, 2009 that it would acquire Barclays Global Investors (BGI), including iShares, from Barclays Bank PLC. The deal closed on December 1, 2009. Consequently, BlackRock became the world's largest investment management company. Observing the news of the BGI transaction on June 11, sophisticated investors would expect the iShares assets under management (AUM) to expand in the following years. For the component stocks of iShares ETFs, these are future demand shocks that are not related to their firm fundamentals. Some other examples may include the future liquidity crisis, sentiment, or behavioral bias. In financial markets, high-frequency traders (HFTs) often learn about future demand shocks.

This article contributes to the literature on rational expectations models with asymmetric information. Our understanding of endogenous information acquisition started from static descriptions about financial markets where signals are only on current fundamentals (Grossman and Stiglitz, 1980; Hellwig, 1980; Verrecchia, 1982): information acquisition by informed investors exhibits strategic substitution. On the one hand, they can gather private information about the liquidation value or dividend to make better portfolio choices, collectively making price more informative. On the other hand, when facing a more informative price, investors would rely less on private signals and more on price, which reduces their incentive to acquire private information. Additional mechanisms are at work

when we consider a more complicated information structure. For example, when different investors are informed of different fundamentals of the asset payoff, trading aggressively on information about one fundamental reduces uncertainty of other fundamentals other investors face, encouraging those investors to learn more about those fundamentals (Goldstein and Yang, 2015). Goldstein and Yang (2019) further specifies dimensions of fundamental uncertainty to be one that is related to the aggregate economy and one that is firm-specific. Disclosing firm-specific information that the real decision maker already knows improves price informativeness and real efficiency. Learning about current demand shocks plays a slightly different role. In Medrano and Vives (2004), Ganguli and Yang (2009) and Manzano and Vives (2011), an investor receives an endowment shock, which can be viewed as her private information about the aggregate endowment shock and helps her make more precise inference about fundamentals. Marmora and Rytchkov (2018) show that learning about noise increases price informativeness, and the price can be the most informative when the majority of investors acquire non-fundamental information.

In dynamic settings, interpretations about fundamental and non-fundamental information can be different, and it becomes meaningful to distinguish current and future shocks. However, learning is still limited to future dividend innovations, e.g. Cai (2017) and Avdis (2016). In Cai (2017), learning about the future is achieved through learning about the persistent part of current and past dividends, which can then be thought of as a signal for future dividends. And in Avdis (2016), the equilibrium price contains both dividend and discount-rate information, where discount-rate shocks are modeled as demand shocks. Thus dividend information acquisition offers an extra benefit of revealing non-dividend information about conditional expected returns. This article approaches dynamic information acquisition in a different way where signals about i.i.d. future demand shocks are acquired today. I emphasize the future uncertainty channel where the asset becomes riskier because less dividend information is learned on the aggregate level and an atomic investor is more certain about asset payoff in the second period thanks to the signal

about second-period demand shocks they acquire in the first period.

This article also connects with the macroeconomic literature on news-driven business cycles. As discussed by Beaudry and Portier (2014), "if an agent can properly anticipate a future need, he can gain by trying to preempt the market and invest early as to make good readily available when the predicted needs eventually appear". The front-running channel in this article echoes this general idea. However, the macroeconomic literature usually treats news shocks as signals about future fundamentals. For example, Jaimovich and Rebelo (2009) shows how news about future total factor productivity affects current output, consumption and investment. This article complements this literature by highlighting the importance of learning about future demand shocks.

This article also contributes to the literature on HFT[1]. In reality, HFTs actively manage their limit orders and reveal information through quote updates (Hasbrouck and Saar, 2013). Empirical evidence also shows that HFTs predict order flows (Hirschey, 2018) or price changes (Harris and Saad, 2014), which can be mapped into acquiring a signal for future demand shocks in this article. This article also finds that with access to information about future demand shocks, investors are better off, which is consistent with Jovanovic and Menkveld (2017) who also find a moderately positive welfare effect.

The rest of this article is organized as follows. Section 2 sets up the model and presents its solutions. Section 3 presents the analytical and numerical results from the static model. Section 4 discusses the empirical applications of the dynamic model, and Section 5 concludes.

## 2. The Model

### *2.1. Setup*

#### *2.1.1. Assets*

There is a single tradable, risky asset, with supply $\bar{x}_t$. To disentangle the future uncertainty and the front-running channels, I introduce a parameter $\alpha \in \{0, 1\}$ as in

[1]Please refer to Menkveld (2016) for a more thorough literature review on HFT.

Farboodi and Veldkamp (2020) and model the payoff of the risky asset as $\alpha p_{t+1} + \tilde{d}_t$, the sum of the asset price and the dividend paid out at the end of period $t$.

When $\alpha = 1$, the risky asset pays $p_{t+1} + \tilde{d}_t$, the future price of the long-lived asset plus its dividend. The dividend is modeled as $\tilde{d}_t = \mu + g d_{t-1} + \tilde{y}_t$, where $\mu$ and $g$ are known parameters, and the dividend shock $\tilde{y}_t \sim i.i.d.\ N(0, \tau_y^{-1})$. In this case, future demand shocks will affect future equilibrium prices, and thus affect reselling profits and current choices. I call this the "dynamic" model.

When $\alpha = 0$, the asset is not long-lived, which can be reinterpreted as a series of dividend strips. That is, in each period, there is a *new* one-period risky asset that can be traded. I call this the "static" model. Studying the static model is important. Firstly, if demand shocks can be insured or hedged, then the static model and the dynamic model are equivalent up to constants and multipliers.[2] Thus the static model can be interpreted as equivalent to a model with a long-lived risky asset where demand shocks can be fully insured or fully hedged. Secondly, using dividend strips allows isolation of the new future uncertainty channel. In the static model, the front-running channel is shut down, but signals about future demand shocks can still transmit via the future uncertainty channel. Thirdly, studying dividend strips per se is also important since they provide information about how stock prices are formed[3].

### *2.1.2. Signals*

Each investor receives an unbiased signal $\tilde{\eta}_{it}^y = \tilde{y}_t + \tilde{\epsilon}_{it}^y$ about current dividend innovation $\tilde{y}_t$, where $\tilde{\epsilon}_{it}^y \sim N(0, \tau_{\epsilon y t}^{-1})$, and an unbiased signal $\tilde{\eta}_{it}^x = \tilde{x}_t + \tilde{\epsilon}_{it}^x$ about current demand shocks $\tilde{x}_t$, where $\tilde{x}_t \sim iid.\ N(0, \tau_x^{-1})$ and $\tilde{\epsilon}_{it}^x \sim N(0, \tau_{\epsilon x t}^{-1})$. Investors can also predict demand shocks

[2]Consider an insurance company. $\mathcal{G}_t$-investor $i$ pays an insurance fee at the beginning of their life, and receives $f_t \mid \mathcal{I}_t^+$ (measurable at the end of period $t$) at the end of period $t$ and $f_{t+1} \mid \mathcal{I}_{t+1}^+$ at the end of period $t+1$, where $\tilde{f}_t$ is the fundamental value of this long-lived asset, and $\tilde{f}_t \mid \mathcal{I}_t^+ = r^2\mu/[(r-1)(r-g)] + rgd_{t-1}/(r-g) + ry_t/(r-g)$. Thus the budget constraint is $\tilde{W}_{i,t+1^+} = r[rW_{it-1^+} + q'_{it}(\tilde{f}_t - p_t)] + q'_{it+1}(\tilde{f}_{t+1} - p_{t+1}r)$. For investor $i$, current and future demand shocks are insured. The aforementioned model is different from the static model only by some constants and multipliers.

[3]For example, van Binsbergen et al. (2012) empirically measure dividend strip prices to study the term structure of the equity premium.

in the next period $\tilde{x}_{t+1}$ by an unbiased signal $\tilde{\xi}^x_{it} = \tilde{x}_{t+1} + \tilde{\nu}^x_{it}$, where $\tilde{\nu}^x_{it} \sim i.i.d.\ N(0, \tau^{-1}_{\nu xt})$. All shocks are orthogonal. Signal precisions[4] $\tau_{\epsilon xt}, \tau_{\epsilon yt}$, and $\tau_{\nu xt}$ are choice variables and investors optimally allocate them with a finite information capacity constraint $K$.

### *2.1.3. Investors*

At the start of each period $t$, a continuum of measure-$\frac{1}{2}$ investors, denoted by $\mathcal{G}_t$, with initial investable wealth $W_{it-1^+} = W_0$, are born. They live for two periods[5], making both information and portfolio choices in each period. Investor $i$ only consumes at the terminal date, and the utility function over the terminal wealth is $-\exp(-\rho W_{it+1^+})$, where $\rho$ is the risk aversion coefficient.

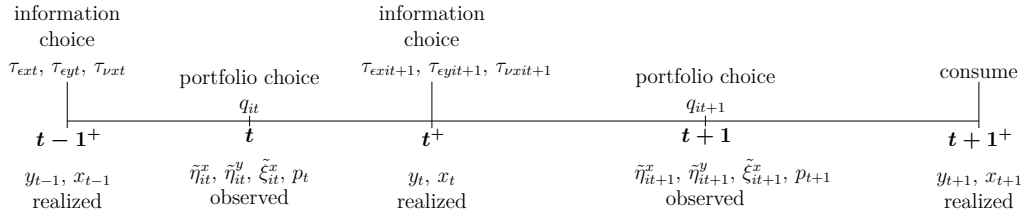



**Fig. 1. Life cycle of investors born at the start of period *t*.** Generation-$t$ investors are born at the beginning of period $t$ (marked as $t-1^+$) and live for two periods: period $t$ and period $t+1$. In period $t$, investors first observe realized past dividend innovations $y_{t-1}$ and demand shocks $x_{t-1}$, then make information choices $\tau_{\epsilon xt}, \tau_{\epsilon yt}, \tau_{\nu xt}$. After receiving private signals $\tilde{\eta}^x_{it}$, $\tilde{\eta}^y_{it}$ and observing the market-clearing price $p_t$, investors choose the optimal portfolio holding $q_{it}$. At the end of period $t$ (denoted $t^+$), the period-$t$ dividend innovation $y_t$ and demand shock $x_t$ are realized. Investors repeat information acquisition and portfolio decisions in period $t+1$, and consume all terminal wealth at $t+1^+$.

At the portfolio choice stage in period $t+1$, after receiving private signals and observing the equilibrium price, $\mathcal{G}_t$-investor $i$ maximizes the expected utility conditional on her available information by choosing how much to invest in the risky asset. That is, investor $i$ chooses the optimal portfolio $\hat{q}_{it+1}$ to maximize her interim expected utility $\mathbb{E}[-\exp(-\rho \tilde{W}_{it+1^+}) \mid \hat{\mathcal{I}}_{it+1}]$

[4]The subscript $i$ is dropped because investors are ex ante identical.

[5]Several papers also model two trading periods, e.g. Holden and Subrahmanyam (2002), Chordia and Subrahmanyam (2004) and Avdis (2016), among others.

subject to the budget constraint[6] :

$$\tilde{W}_{it+1^+} = rW_{i,t^+} + \hat{q}_{it+1}(\tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1}r). \tag{1}$$

At the beginning of period $t+1$, investor $i$ chooses signal precision $\hat{\tau}_{\epsilon yit+1}$, $\hat{\tau}_{\epsilon xit+1}$, and $\hat{\tau}_{\nu xit+1}$ to maximize expected utility $\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it+1^+}) \;\big|\; \hat{\mathcal{I}}_{it}^{+}\big]$ subject to the information capacity constraint

$$\gamma_{\epsilon y}\hat{\tau}_{\epsilon yit+1} + \gamma_{\epsilon x}\hat{\tau}_{\epsilon xit+1} + \gamma_{\nu x}\hat{\tau}_{\nu xit+1} \leq K. \tag{2}$$

The information constraint is assumed to be linear in signal precision, that is the marginal cost of an extra unit of signal precision is the same at all signal precision levels.[7] This assumption helps to isolate the benefits of learning about future demand shocks, i.e. the future uncertainty channel, from a cost reduction channel. Another popular choice of information constraint in the literature is a convex one, so that the marginal cost of learning within one signal is increasing as investors allocate more precision to that signal. In that case, the possibility to learn about future demand shocks today and tomorrow makes learning about demand shocks effectively cheaper, because investors can smooth learning costs. This cost reduction channel works in the same direction as the benefit increase through the future uncertainty channel, and will make the results even stronger. For the purpose of demonstrating the future uncertainty channel clearly, here I use a linear constraint.

When making portfolio choices in period $t$, $\mathcal{G}_t$-investor $i$ also has to take into consideration future choices in period $t+1$ and thus chooses $q_{it}$ and ${q_{it+1}}$[8] to maximize the expected utility

[6]Throughout this article, the notation $\hat{}$ is used on variables that are related to the old generation.

[7]The linear information capacity constraint is common in the literature; see Kacperczyk et al. (2016) among others.

[8]Technically, $\mathcal{G}_t$-investor $i$ also needs to include three other choice variables, namely $\tau_{\epsilon xit+1}$, $\tau_{\epsilon yit+1}$ and $\tau_{\nu xit+1}$. They can be omitted, however, for two reasons. Firstly, we only solve portfolio choices in a functional form of conditional expectations and variances of future payoffs. Although those moments depend on the optimal information choices in period $t+1$, that functional form does not. Secondly, solving for the optimal information choices in period $t+1$ is also useless since the investor will make better information choices after $d_t$ is realized at the beginning of period $t+1$. The same reasoning goes for the information choice problem at the beginning of period $t$.

$\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it+1^+}) \,\big|\, \mathcal{I}_{it}\big]$, subject to the budget constraint

$$\tilde{W}_{it+1^+} = r^2 W_0 + r q_{it}(\tilde{d}_t + \alpha p_{t+1} - p_t r) + \hat{q}_{it+1}(\tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1} r). \tag{3}$$

Notice the difference between $\hat{q}_{it+1}$ and $q_{it+1}$. While $\hat{q}_{it+1}$ is the actual investment that $\mathcal{G}_t$-investor $i$ makes at the portfolio choice stage of period $t+1$, based on all previous dividends and signal realizations, $\hat{q}_{it+1}$ is what $\mathcal{G}_t$-investor $i$ predicts herself to hold in period $t$ based on only the information that is available to her at the portfolio choice stage in period $t$.

Finally, in the beginning of period $t$, $\mathcal{G}_t$-investor $i$ makes period-$t$ information choices $\tau_{\epsilon y t}$, $\tau_{\epsilon x t}$, and $\tau_{\nu x t}$ to maximize her ex ante utility $\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it+1^+}) \,\big|\, \mathcal{I}^+_{t-1}\big]$ subject to the information capacity constraint

$$\gamma_{\epsilon y}\tau_{\epsilon y t} + \gamma_{\epsilon x}\tau_{\epsilon x t} + \gamma_{\nu x}\tau_{\nu x t} \leq K. \tag{4}$$

*2.1.4. Information structure*

The equilibrium price is a linear function of the signals. Since signals are unbiased, the price can be expressed as a linear function of shocks:

$$p_t = A_t + B_t d_{t-1} + C_t \tilde{y}_t + D_t \tilde{x}_t + E_t \tilde{x}_{t+1}. \tag{5}$$

This is different from existing literature in that there is an extra term of the future demand shock, $\tilde{x}_{t+1}$, in the equilibrium price. For future demand shocks to enter the equilibrium price relation, we need two conditions. First, signals about future demand shocks must exist and be available for acquisition. Otherwise, as in Farboodi and Veldkamp (2020) which modeled a long-lived asset and signals about current dividend and demand shocks, the future demand shock $\tilde{x}_{t+1}$ does not enter the equilibrium price relation. Second, the asset has to be long-lived ($\alpha = 1$). Otherwise when $\alpha = 0$, $\tilde{d}_t$ and $\tilde{d}_{t+1}$ are different assets, and the demand shocks associated with $\tilde{d}_{t+1}$ do not affect the equilibrium price of $\tilde{d}_t$ directly, i.e. $\tilde{x}_{t+1}$ does not appear

in the equilibrium price, yet it could indirectly make a difference through price coefficients due to information acquisition.

Define the price-to-current-noise ratio to be $\zeta_{1t} \equiv C_t/D_t$, and the price-to-future noise ratio to be $\zeta_{2t} \equiv C_t/E_t$, where $C_t$ is the price sensitivity to dividend innovation, and $D_t$ and $E_t$ are the price sensitivities to current and future demand shocks, respectively, in the equilibrium price (5). A transformation of the equilibrium price $p_t$ can generate an unbiased signal about current dividend shock. The two generations observe the same price but extract different information from it about $\tilde{y}_t$. In period $t$, $\mathcal{G}_t$-investor $i$ extracts a signal $\tilde{\eta}^p_{it} = \tilde{y}_t + \tilde{\nu}^p_{it}$, where $\tilde{\nu}^p_{it} = \big(\tilde{x}_t - \mathbb{E}\big[\tilde{x}_t \mid \tilde{\eta}^x_{it}\big]\big)/\zeta_{1t} + \big(\tilde{x}_{t+1} - \mathbb{E}\big[\tilde{x}_{t+1} \mid \tilde{\xi}^x_{it}\big]\big)/\zeta_{2t} \sim N\big(0, \tau^{-1}_{pt}\big)$, with the variance of the price signal being $\tau^{-1}_{pt} = \zeta^{-2}_{1t} V_{xt} + \zeta^{-2}_{2t} V^{\xi}_{xt}$.[9] $\mathcal{G}_{t-1}$-investor $i$ extracts a signal $\tilde{\hat{\eta}}^p_{it} = \tilde{y}_t + \tilde{\hat{\nu}}^p_{it}$, where $\tilde{\hat{\nu}}^p_{it} \equiv \big(\tilde{x}_t - \mathbb{E}\big[\tilde{x}_t \mid \tilde{\eta}^x_{it}, \tilde{\xi}^x_{it-1}\big]\big)/\zeta_{1t} + \big(\tilde{x}_{t+1} - \mathbb{E}\big[\tilde{x}_{t+1} \mid \tilde{\hat{\xi}}^x_{it}\big]\big)/\zeta_{2t} \sim N\big(0, \hat{\tau}^{-1}_{pit}\big)$, with the variance of the price signal being $\hat{\tau}^{-1}_{pit} = \zeta^{-2}_{1t} \hat{V}_{xit} + \zeta^{-2}_{2t} \hat{V}^{\xi}_{xit}$. To proceed, this article defines an important concept of *price informativeness*.

Definition 1. *Price informativeness for investor $i$ is defined by the precision of the price signal given the dividend realization and the investor's information about demand shocks. Mathematically, in period $t$, the price informativeness is $\mathbb{V}ar^{-1}\big(\tilde{\nu}^p_{it} \mid \tilde{\eta}^x_{it}, \tilde{\xi}^x_{it}\big)$ for $\mathcal{G}_t$ and $\mathbb{V}ar^{-1}\big(\tilde{\hat{\nu}}^p_{it} \mid \tilde{\eta}^x_{it}, \tilde{\xi}^x_{it-1}, \tilde{\hat{\xi}}^x_{it}\big)$ for $\mathcal{G}_{t-1}$ investors.*

To summarize, each investor processes all available information and obtains two signals – the dividend signal and the price signal – concerning the dividend innovation. Private knowledge about $\tilde{x}_t$ affects their ability to extract information about the dividend innovation from the equilibrium price. In other words, it affects price informativeness. Investors can learn more from the price either when they learn more about $\tilde{x}_t$ so that they can better extract information from the price, or other investors have learned more about $\tilde{x}_t$ so that there is not much noise contained in the price. The acquired signal $\tilde{\xi}^x_{it}$ about future demand shocks will affect next period's learning abilities and the marginal value of learning.

[9] $V_{xt}$, $V^{\xi}_{xt}$, $\hat{V}_{xit}$ and $\hat{V}^{\xi}_{xit}$ are defined in the appendix. The subscript $i$ is dropped from them and $\tau_{pt}$ because in period $t$, $\mathcal{G}_t$-investors are ex ante identical.

Let $\mathcal{F}_{t-1}^{+}$ be the information set that contains all past realizations of dividend innovations, demand shocks and equilibrium prices. The two generations have different information sets at different choice stages, which can be summarized in Table 1.

**Table 1.** Information sets of the two generations in Period $t$

| Generation | Information Choice Stage | Portfolio Choice Stage |
|---|---|---|
| $\mathcal{G}_t$ | $\mathcal{I}_{t-1}^{+} = \mathcal{F}_{t-1}^{+}$ | $\mathcal{I}_{it} = \left\{\mathcal{F}_{t-1}^{+}, \eta_{it}^{y}, \eta_{it}^{x}, \xi_{it}^{x}, p_t\right\}$ |
| $\mathcal{G}_{t-1}$ | $\hat{\mathcal{I}}_{it-1}^{+} = \left\{\mathcal{F}_{t-1}^{+}, \xi_{it-1}^{x}\right\}$ | $\hat{\mathcal{I}}_{it} = \left\{\mathcal{F}_{t-1}^{+}, \xi_{it-1}^{x}, \hat{\eta}_{it}^{y}, \hat{\eta}_{it}^{x}, \hat{\xi}_{it}^{x}, p_t\right\}$ |

$\mathcal{G}_t$-investors and $\mathcal{G}_{t-1}$-investors are endowed with heterogeneous information sets. Specifically, the knowledge regarding $\tilde{x}_t$ acquired by $\mathcal{G}_{t-1}$-investors in period $t-1$ persists and exerts influence on their optimal decisions in the subsequent period $t$.

$\mathcal{F}_{t-1}^{+}$ can be interpreted as what an outsider knows about the market at the beginning of period $t$. Ex ante, $\mathcal{G}_t$-investors know as much as an outsider. At the portfolio choice stage, however, they have heterogeneous information due to received private signals. After the current dividend innovation is realized, signals about the current dividend innovation and demand shock no longer assist in future decision making. Only the signal realization about the next period's demand shock persists and affects future choices.

#### *2.1.5. Benchmark model*

The benchmark, corresponding to the standard specification in the existing literature, shuts down acquisition of information about future demand shocks. It is denoted by the superscript $B$ in this article. If the precision of the future demand signal $\tau_{\nu x}$ is zero, or the cost of learning about future demand shocks $\gamma_{\nu x}$ is infinitely large, the main model collapses to the benchmark model.

### *2.2. Market Clearing and Equilibrium*

This article introduces acquisition of information about future demand shocks to the noisy rational expectations equilibrium. See Appendix A for a formal definition. The risky

asset price clears the market when the aggregate demand equals the aggregate supply:

$$\int_{i\in\mathcal{G}_t} q_{it}\,\mathrm{d}i + \int_{i\in\mathcal{G}_{t-1}} \hat{q}_{it} - q_{it-1}\,\mathrm{d}i = \bar{x}_t + \tilde{x}_t + \int_{i\in\mathcal{G}_{t-2}} \hat{q}_{it-1}\,\mathrm{d}i\,. \tag{6}$$

*2.3. Model solutions*

*2.3.1. Portfolio choices*

For notational simplicity, denote the expected period-$t$ payoff and the expected period-$(t+1)$ payoff conditional on $\mathcal{G}_t$-investor $i$'s interim information set in period $t$ as $\mathcal{E}_{it} \equiv \mathbb{E}[\tilde{d}_t + \alpha p_{t+1} \mid \mathcal{I}_{it}] - p_t r$ and $\mathcal{E}_{2i} \equiv \mathbb{E}[\tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1} r \mid \mathcal{I}_{it}]$, respectively, and their variances as $\mathcal{V}_t \equiv \mathbb{V}ar(\tilde{d}_t + \alpha p_{t+1} \mid \mathcal{I}_{it})$ and $\mathcal{V}_2 \equiv \mathbb{V}ar(\tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1} r \mid \mathcal{I}_{it})$, respectively. The covariance of the period-$t$ payoff and the period-$(t+1)$ payoff conditional on $\mathcal{I}_{it}$ is denoted as $\mathcal{C} \equiv \mathbb{C}ov(\tilde{d}_t + \alpha p_{t+1}, \tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1} r \mid \mathcal{I}_{it})$. These conditional moments are derived in Appendix B.1.1 and Appendix B.1.2. The portfolio choice of $\mathcal{G}_t$-investor $i$ can then be written as

$$q_{it} = \frac{1}{\rho r}\left(\frac{\mathcal{V}_2}{\mathcal{V}_t\mathcal{V}_2 - \mathcal{C}^2}\mathcal{E}_{it} - \frac{\mathcal{C}}{\mathcal{V}_t\mathcal{V}_2 - \mathcal{C}^2}\mathcal{E}_{2i}\right). \tag{7}$$

If the conditional expectation about future excess payoffs ($\mathcal{E}_{2i}$) changes, the individual demand curve undergoes a parallel shift. But when the conditional variance of future excess payoffs becomes smaller or the conditional covariance of current and future excess payoffs becomes larger, the slope of the demand curve would become less steep.

$\mathcal{G}_{t-1}$-investor $i$ faces a simple one-shot portfolio choice problem, which has an optimal solution

$$\hat{q}_{it} = \frac{\hat{\mathcal{E}}_{it}}{\rho\hat{\mathcal{V}}_{it}}, \tag{8}$$

where $\hat{\mathcal{E}}_{it} \equiv \mathbb{E}[\tilde{d}_t + \alpha p_{t+1} \mid \hat{\mathcal{I}}_{it}] - p_t r$ and $\hat{\mathcal{V}}_{it} \equiv \mathbb{V}ar(\tilde{d}_t + \alpha p_{t+1} \mid \hat{\mathcal{I}}_{it})$.

*Static case.* In the static case where $\alpha = 0$, no portfolio rebalance is involved. There is no generation difference and each period $t$ is an identical snapshot from the timeline. It is worth noticing is that the covariance term $\mathcal{C}_t = 0$. The reasoning is as follows. Conditional on $\mathcal{I}_{it}$,

the random components in $\tilde{d}_t - p_t$ is $\tilde{y}_t \mid \tilde{\eta}^y_{it}, \tilde{\eta}^p_{it}$ and in $\tilde{d}_{t+1} - p_{t+1}$ are $\tilde{y}_{t+1}$, $\tilde{x}_{t+1} \mid \tilde{\xi}^x_{it}$ and $\tilde{x}_{t+2}$, respectively. They are orthogonal (see the proof of Lemma 4 in the appendix). This effectively separates the two portfolio choices of investors. Therefore, the optimal portfolio choices for $\mathcal{G}_t$- and $\mathcal{G}_{t-1}$-investors are

$$q_{it} = \frac{\mathbb{E}\big[\tilde{d}_t \mid \mathcal{I}_{it}\big] - p_t r}{\rho r \mathbb{V}ar\big(\tilde{d}_t \mid \mathcal{I}_{it}\big)} \quad \text{and} \quad \hat{q}_{it} = \frac{\mathbb{E}\big[\tilde{d}_t \mid \hat{\mathcal{I}}_{it}\big] - p_t r}{\rho \mathbb{V}ar\big(\tilde{d}_t \mid \hat{\mathcal{I}}_{it}\big)},$$

respectively.

*2.3.2. Equilibrium price relation*

For simplicity, define $\bar{\chi}_{yt} \equiv \Omega_t V_{yt}/r + \int_{i\in\mathcal{G}_{t-1}} \hat{V}_{yit}/\hat{\mathcal{V}}_{it}\,\mathrm{d}i$ and $\bar{\chi}_{xt} \equiv \Omega_t V_{xt}/r + \int_{i\in\mathcal{G}_{t-1}} \hat{V}_{xit}/\hat{\mathcal{V}}_{it}\,\mathrm{d}i$ (refer to Appendix B.2 for details). In equilibrium, the period-$t$ price coefficients are recursively characterized by

$$\begin{aligned} A_t = {} & \left(1 - \frac{\bar{\Sigma}_t}{2r}\frac{\mathcal{C}}{\mathcal{V}_t\mathcal{V}_2 - \mathcal{C}^2}\right)\frac{\mu}{r-\alpha g} - \frac{\bar{\Sigma}_t\rho}{r}\bar{x}_t + \frac{1}{r}\left(\alpha + \frac{\bar{\Sigma}_t}{2r}\frac{\mathcal{C}}{\mathcal{V}_t\mathcal{V}_2 - \mathcal{C}^2}\right)A_{t+1} \\ & - \frac{\bar{\Sigma}_t}{2r^2}\frac{\mathcal{C}}{\mathcal{V}_t\mathcal{V}_2 - \mathcal{C}^2}\alpha A_{t+2}, \end{aligned} \tag{9}$$

$$B_t = \frac{g}{r-\alpha g}, \tag{10}$$

$$C_t = \frac{1}{r-\alpha g}\left(1 - \bar{\Sigma}_t\bar{\chi}_{yt}\tau_y\right), \tag{11}$$

$$\begin{aligned} D_t = {} & -\frac{\rho\bar{\Sigma}_t}{r} + \frac{1}{r-\alpha g}\bar{\Sigma}_t\bar{\zeta}^{\dagger}_{1t}\tau_x, \\ E_t = {} & \frac{1}{r}\left[\frac{\bar{\Sigma}_t}{2r}\frac{\mathcal{C}}{\mathcal{V}_t\mathcal{V}_2 - \mathcal{C}^2}V^{\xi}_{xt}\tau_{\nu xt} + \alpha\left(1 - \bar{\Sigma}_t\bar{\chi}_{xt}\tau_x\right)\right]D_{t+1} + \frac{1}{r-\alpha g}\bar{\Sigma}_t\bar{\zeta}^{\dagger}_{2t}\tau_x. \end{aligned} \tag{12}$$

$B_t$ is the constant dividend growth rate, and governs the level of the equilibrium price together with $A_t$. $C_t$ is the price sensitivity to dividend innovation. $D_t$ and $E_t$ regulate the amount of current and future demand shocks contained in the price, respectively. $\bar{\zeta}^{\dagger}_{1t}$ and $\bar{\zeta}^{\dagger}_{2t}$ are the market-wide average future-uncertainty-adjusted dividend-to-total-uncertainty-weighted signal-to-current-noise and signal-to-future-noise

ratios, defined by (B.43) and (B.44), respectively.

*Static case.* When $\alpha = 0$, the equilibrium price relation greatly simplifies. Firstly, $E_t = 0$. This means that here $\tilde{x}_{t+1}$ does not directly enter the equilibrium price at $t$. This is because $\tilde{d}_t$ and $\tilde{d}_{t+1}$ are different assets, and the demand shock to $\tilde{d}_{t+1}$ does not affect the equilibrium price of $\tilde{d}_t$ directly, but only indirectly though price coefficients due to information acquisition. Secondly, $\mathcal{C}_t = 0$, that is the current and future excess payoffs are conditionally independent. Thirdly, $\mathcal{G}_{t-1}$-investors make homogeneous period-$t$ information choices. The price coefficients then become

$$\begin{aligned} A_t &= \frac{1}{r}\left(\mu - \bar{\Sigma}_t \rho \bar{x}_t\right), \quad B_t = \frac{g}{r}, \quad C_t = \frac{1}{r}\left[1 - \frac{r+1}{2r}\,\bar{\Sigma}_t \tau_y\right], \\ D_t &= \frac{1}{r}\left[-\rho\bar{\Sigma}_t + \frac{r+1}{2r}\,\bar{\Sigma}_t \zeta_t \tau_x\right]. \end{aligned} \tag{13}$$

*2.3.3. Information choices*

*Ex ante expected utilities.* In period $t$, $\mathcal{G}_{t-1}$-investors are in the second period of their life. The objective function is $\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it^+}) \mid \hat{\mathcal{I}}^+_{it-1}\big]$ and the constraint of the information choice problem is very similar to (2), with only a change of subscript $t$. This yields an objective for investor $i \in \{\mathcal{G}_{t-1}\}$ that is increasing and concave in the precisions of the fundamental signal and the current demand signal:

$$\begin{aligned} \mathbb{E}\big[-\exp\big(-\rho r W_{it^+}\big) \,\Big|\, \hat{\mathcal{I}}_{it-1^+}\big] = -\big|\hat{\Sigma}_{t-1^+}/\hat{\mathcal{V}}_{it}\big|^{-1/2} \times \\ \exp\big(-\rho r W_{it-1^+}\big) \exp\left(-\frac{\mu^2_{\hat{\mathcal{E}}_t}}{2\hat{\Sigma}_{t-1^+}}\right). \end{aligned} \tag{14}$$

Here $\mu_{\hat{\mathcal{E}}_t}$ and $\hat{\Sigma}_{t-1^+}$ are defined by (B.18) and (B.17), respectively. Notice that in the static model, investors optimally choose not to learn about future demand shocks. Thus $\hat{\tau}_{\nu x i t+1}$ does not affect the ex ante utility.

For $\mathcal{G}_t$-investors, the information choice problem is to maximize $\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it+1^+}) \mid \mathcal{I}^+_{t-1}\big]$ subject to (4). This yields an objective for $i \in \{\mathcal{G}_t\}$ that is increasing and concave in

precisions of the signals of the current dividend innovation, the current demand shock and the future demand shock:

$$\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right)\middle|\mathcal{I}_{t-1}^{+}\right] = -\exp\left(-\rho r^2 W_0\right)\times \\ |I-2\Sigma F|^{-\frac{1}{2}}\times\sqrt{\hat{\mathcal{V}}_{it+1}/\left(\hat{\mathcal{V}}_{it+1}+\Sigma_{\hat{Z}_{t+1}}\right)}\times \\ \exp\left(\tfrac{1}{2}G'\left(I-2\Sigma F\right)^{-1}\Sigma G+J\right), \tag{15}$$

where $\hat{\mathcal{V}}_{it+1}$, $\Sigma_{\hat{Z}_{t+1}}$, $\Sigma$, $F$, $G$ and $J$ are defined in the appendix.

*Static case.* Information choices in the dynamic model are very complex. To develop intuition, here I derive the analytical solution of information choices in the static model. To understand how investors would allocate precision to signals about the current dividend innovation and the current demand shock, it is convenient to first define the concept of learning index.

Definition 2. *The learning index for a signal is defined as the increase in ex ante marginal utility associated with an extra unit of precision assigned to this signal.*

For $\mathcal{G}_{t-1}$-investors, the learning index for the signals about the current dividend innovation and the current demand shock are $\partial\widehat{EU}/\partial\hat{\tau}_{\epsilon yit}$ and $\partial\widehat{EU}/\partial\hat{\tau}_{\epsilon xit}$, respectively. The learning indices are functions of current precision choices $\hat{\tau}_{\epsilon yit}$ and $\hat{\tau}_{\epsilon xit}$, the signal precision $\tau_{\nu xt-1}$, and the realization of the signal about current demand shocks received in the previous period, $\xi_{it-1}^{x}$. Since all investors are ex ante identical, the signal precision $\tau_{\nu xt-1}$ is the same across $\mathcal{G}_{t-1}$-investors. However, signal realizations $\xi_{it-1}^{x}$ are private information to investors, introducing heterogeneity in second-period learning indices. The effects of $\xi_{it-1}^{x}$ on learning indices for signals about the current dividend innovation and the current demand shock affect both learning indices proportionally, so their ratio does not depend on signal realizations. This observation can be summarized by the following lemma.

LEMMA 1 (Proportionality of second-period learning indices). *The ratio of second-period learning indices for the current dividend innovation and the current demand shock is* $\hat{L}_{\epsilon xit}/\hat{L}_{\epsilon yit} = \zeta_t^2$.

To better characterize the benefit of learning about future demand shocks, it is useful to study it when learning about different shocks is equally costly. The equalcost assumption is stated below for later reference.

ASSUMPTION 1. *Acquiring signals about current dividend innovation, current and future demand shocks is equally costly, that is* $\gamma_{\epsilon y} = \gamma_{\epsilon x} = \gamma_{\nu x}$.

The information choices of $\mathcal{G}_{t-1}$-investors can then be summarized by the following proposition.

PROPOSITION 1 (Second-period information choice). *In period* $t$, $\mathcal{G}_{t-1}$*-investors*

*(a) optimally choose not to learn about future demand shocks, that is* $\hat{\tau}_{\nu xit} = 0$*; and*

*(b) [under Assumption 1] only learn about current dividend innovation regardless of the realization of signal about* $\tilde{x}_t$ *they received in the previous period, that is* $\hat{\tau}_{\epsilon yit} = K/\gamma_{\epsilon y}$ *and* $\hat{\tau}_{\epsilon xit} = 0$.

What $\mathcal{G}_{t-1}$-investors have learned in the past about today does not affect their information choice for today. It should be noted, however, that portfolio choices and payoffs are affected. This arises because they do have an additional signal about today's asset. Their conditional mean and variance about today's asset are different from those of $\mathcal{G}_t$-investors.

In the first period, the proportionality of learning indices for signals about the current dividend innovation and the current shock persists, as is summarized in Lemma 2.

LEMMA 2 (Proportionality of first-period learning indices). *The ratio of first-period learning indices for the current dividend innovation and the current demand shock only depends on the signal-to-noise ratio, that is* $L_{\epsilon xit}/L_{\epsilon yit} = \zeta_t^2$.

The first-period information choices can be summarized by Proposition 2.

PROPOSITION 2 (First-period information choice). *Under Assumption 1, $\mathcal{G}_t$-investors optimally choose to only learn about future demand shocks, that is $\tau_{\epsilon yt} = \tau_{\epsilon xt} = 0$ and $\tau_{\nu xt} = K/\gamma_{\nu x}$.*

Because $\mathcal{G}_t$-investors are identical at the information choice stage of period $t$, they will make the same choice of precision, and thus will have the same learning indices. Compared with the benchmark with no future demand shocks, the learning indices have changed because the option to learn about future demand shocks affects $\bar{\Sigma}_t$, the across-generation average of posterior variance of $\tilde{y}_t$, in two ways. Firstly, $\mathcal{G}_t$-investors optimally choose to only acquire signals about $\tilde{x}_{t+1}$ and do not learn about any current shocks. Secondly, the learning indices of $\mathcal{G}_{t-1}$-investors are affected by their observed private signal $\tilde{\tilde{\xi}}^x_{it}$. Their portfolio choices change accordingly. Changes in learning indices of the old generation spill over to the young generation through $\bar{\Sigma}_t$.

Compared with learning about $\tilde{x}_{t+1}$ in period $t+1$, learning about it in period $t$ has an extra bite: it decreases the variance of future excess payoffs conditional on today's information.

## 3. Results

With a long-lived risky asset, learning about future demand shocks affects the market via two mechanisms: the front-running channel and the future uncertainty channel. To see how the novel future uncertainty channel works more clearly, this section studies the static model and sets $\alpha = 0$ to shut down the front-running channel. In the dynamic model where $\alpha = 1$, the front-running channel works in the same direction as the future uncertainty channel and will strengthen the results in the static model.

### *3.1. Benchmark model*

In the benchmark model, investors do not have access to signals about future demand shocks. The results are shown in Appendix C. When $\gamma_{\nu x} \to \infty$ in the main model,

information capacity constraints make learning future demand shocks prohibitively costly, so investors optimally abstain from acquiring such information. Accordingly, the main model collapses into the benchmark model. When learning future demand shocks is sufficiently costly, the main model converges to the benchmark model.

PROPOSITION 3. *There exists a threshold $\bar{\gamma}_{\nu x}$ such that for all $\gamma_{\nu x} > \bar{\gamma}_{\nu x}$, the results in Proposition 11 hold for the full main model.*

### *3.2. Price reveals less fundamental information*

When we grant investors another source for non-fundamental information, the informational content of the price changes accordingly. We now examine this issue by looking at the signal-to-noise ratio, the average posterior precision about the dividend innovation and price informativeness.

Understanding how the signal-to-noise ratio $|\zeta_t|$ changes with model parameters helps us assess the quality of information in the market. In the benchmark model and the second-period information choice of the main model, $|\zeta_t|$ summarizes all relevant informational content in the market; together with parameters associated with learning costs ($\gamma_{\epsilon y}$, $\gamma_{\epsilon x}$, $\gamma_{\nu x}$ and $K$), the signal-to-noise ratio $|\zeta_t|$ determines how investors allocate signal precision in Benchmark I and in the second period of the main model.

Investors have a prior belief about the dividend innovations. The average posterior precision about the dividend innovation $\bar{\Sigma}_t^{-1}$ is the sum of the prior precision of the dividend innovation, the precision of the dividend signal, and the precision of the price signal. It is a function of the risk-free rate, the risk-aversion coefficient, prior precision of dividend innovation and demand shock, and the signal-to-noise ratio $|\zeta_t|$. $\bar{\Sigma}_t^{-1}$ and $|\zeta_t|$ often comove with changes in parameters, as is shown in Figure 2.

LEMMA 3. *The market-wide average posterior precision for the dividend innovation $\bar{\Sigma}_t^{-1}$ comoves positively with the signal-to-noise ratio $|\zeta_t|$ whenever variations in $|\zeta_t|$ stem from shifts in the parameters $K$, $\gamma_{\epsilon x}$, $\gamma_{\epsilon y}$, $\gamma_{\nu x}$, or $\tau_y$.*

PROPOSITION 4. *With access to information about future demand shock,*

*(a) the signal-to-noise ratio* $|\zeta_t|$ *and the market-wide average posterior precision about the dividend innovation* $\bar{\Sigma}_t^{-1}$ *increase with information processing capacity* $K$ *and prior precision of demand shocks* $\tau_x$ *;*

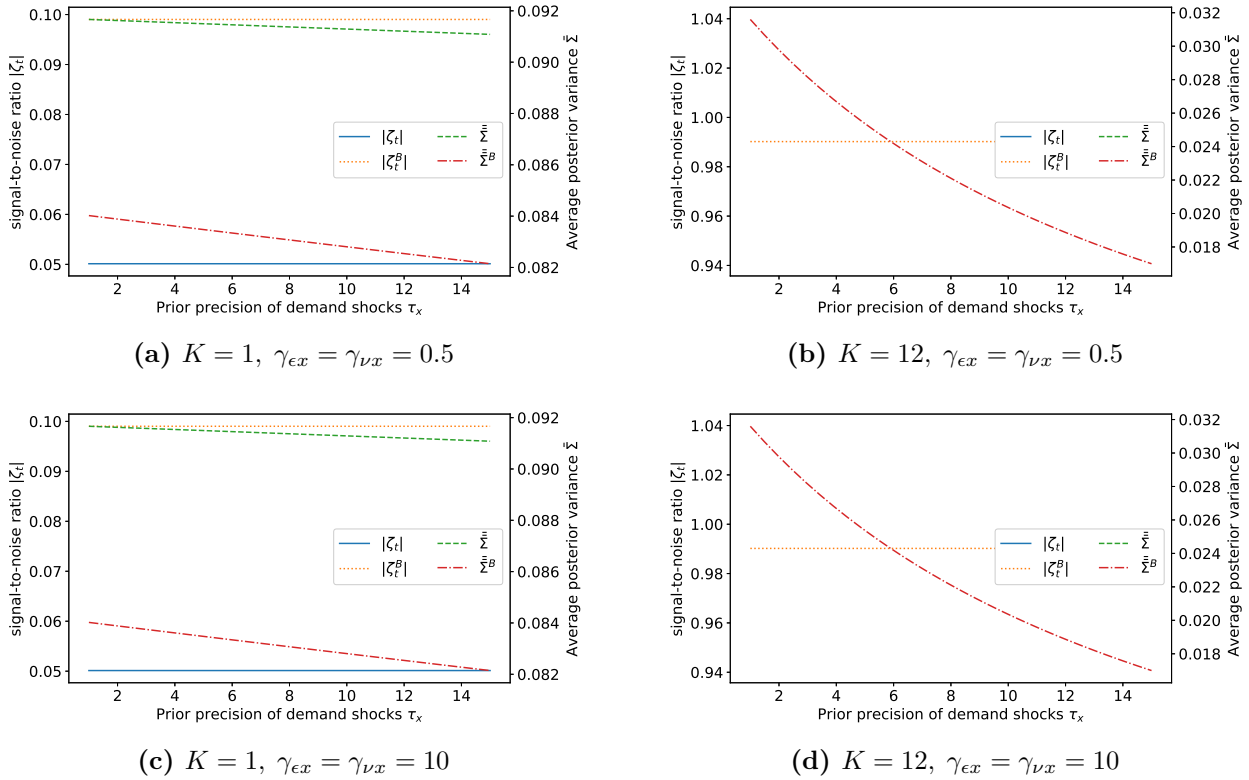
*(b) Assume that the learning costs associated with demand shocks are the same today and tomorrow, i.e.* $\gamma_{\epsilon x} = \gamma_{\nu x} = \gamma_x$*. Then there exists* $\bar{K}$*,* $\gamma_x$ *and* $\gamma_x^*$ *such that when* $K > \bar{K}$ *and* $\gamma_x < \gamma_x$*,* $|\zeta_t|$ *and* $\bar{\Sigma}_t^{-1}$ *decrease with* $\gamma_x$ *if* $\gamma_x < \gamma_x^*$ *and increase with* $\gamma_x$ *if* $\gamma_x > \gamma_x^*$*. When* $K \leq \bar{K}$ *or* $\gamma_x \geq \gamma_x$*,* $|\zeta_t|$ *and* $\bar{\Sigma}_t^{-1}$ *increase with* $\gamma_x$*. Mathematically,*

$$\operatorname{sign} \frac{\partial |\zeta_t|}{\partial \gamma_x} = \operatorname{sign} \frac{\partial \bar{\Sigma}_t^{-1}}{\partial \gamma_x} \begin{cases} -, & \text{if } K > \bar{K},\ \gamma_x < \gamma_x,\ \gamma_x < \gamma_x^*, \\ +, & \text{otherwise.} \end{cases}$$

PROPOSITION 5. *Compared with Benchmark I, the signal-to-noise ratio* $|\zeta_t|$ *decreases when investors have an extra source of information on future demand shocks. The same result applies to the average posterior precision* $\bar{\Sigma}_t^{-1}$*.*

THEOREM 1 (Price informativeness). *With access to information about future demand shocks, price informativeness decreases.*

Figure 3a plots price informativeness against the information processing capacity. For the two periods in the life of investors, the equilibrium price is not equally informative. In the second period, price is more informative because she has acquired a signal about second-period demand shocks in her first period. Although the price contains less information about dividend innovations, her ability to extract other investors' private information out of the equilibrium price is higher, and the latter effect dominates. However, in the first period, the ability to extract information from the price is not changed. Since price contains less fundamental information, it is less informative for her.




**(a)** $K = 1,\ \gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$

**(b)** $K = 12,\ \gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$

**(c)** $K = 1,\ \gamma_{\epsilon x} = \gamma_{\nu x} = 10$

**(d)** $K = 12,\ \gamma_{\epsilon x} = \gamma_{\nu x} = 10$

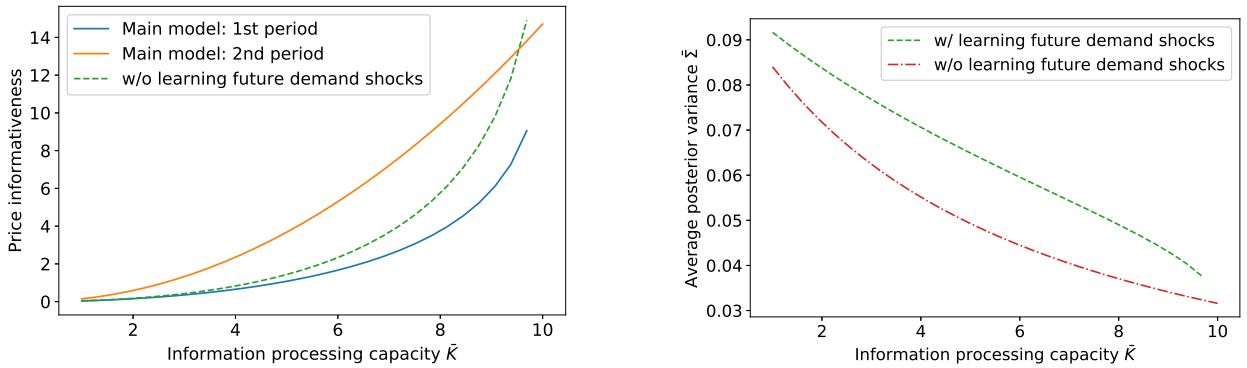
**Fig. 2. The signal-to-noise ratio and the average posterior precision.** These figures plot how the signal-to-noise ratio $|\zeta_t|$ and the average posterior precision $\bar{\Sigma}_t$ change with $K$ and learning costs for demand shocks. Relative to the benchmark model with no acquisition of information about future demand shocks, the equilibrium signal-to-noise ratio declines while the average posterior variance rises. In both models, the signal-to-noise ratio rises and average posterior variance falls as information processing capacity $K$ increases.

Figure 3b plots the average posterior variance of the dividend innovation against the information processing capacity. On the aggregate level, less fundamental information is acquired in the market. As we can see here, the average posterior variance of dividend innovations is larger when investors can learn about future demand shocks. As a result, the risk premium is higher.

### *3.3. Information choices*

THEOREM 2. *When it is possible to learn from future demand shocks, investor i's information choices change in both periods:*

*(a) [**fundamental vs. non-fundamental**] Agent i learns less about the dividend*




**(a)** $K = 1,\ \gamma_{\epsilon x} = \gamma_{\nu x} = 10$ **(b)** $K = 12,\ \gamma_{\epsilon x} = \gamma_{\nu x} = 10$

**Fig. 3. Impact of learning about future demand shocks on price informativeness and risk premium.** Panel (a) plots price informativeness against the information processing capacity. In general, price informativeness rises with information processing capacity. Investors across heterogeneous life stages hold distinct information sets, which generates dispersion in their capacity to recover dividend signals from equilibrium prices. Panel (b) depicts the average posterior variance of dividend innovations as a function of information processing capacity. Because the average posterior variance of dividend innovations is positively associated with the risk premium, the figure illustrates that permitting investors to learn about future demand shocks elevates equilibrium risk premiums.

*innovations today compared with the benchmark;*

(b) *[**current vs. future**] Agent $i$ learns more about the future demand shock than the current demand shock.*

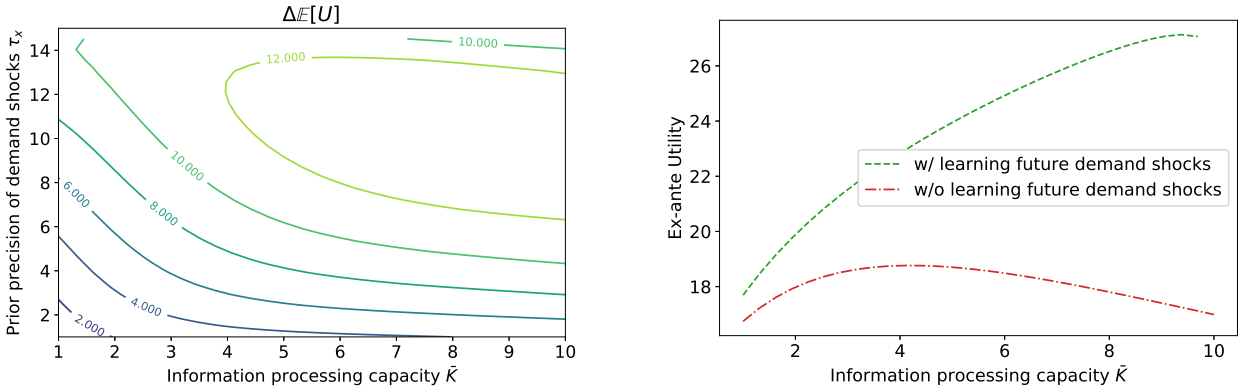
Since the marginal cost of allocating precision to one single signal is increasing, granting access to another signal about demand shocks to investors essentially makes demand shocks cheaper to learn. As a result, investor $i$ will learn less about the dividend innovations today. The precisions of the signals about current and future demand shocks contribute to the conditional mean and variance of $\tilde{x}_t$ in a symmetric way. Starting from the benchmark equilibrium, for each unit of $\tau_{\epsilon xt}$ that investor $i$ gives up, she can allocate more than one unit to $\tau_{\nu xt}$. Since investor $i$ lives for two periods, allocating precision to $\tau_{\nu xt}$ has an extra benefit of lowering the conditional variance of excess payoff from the perspective of today, which in turn affects tomorrow's learning index.

### *3.4. Ex ante utility*

In practice, high-frequency traders are learning about future demand shocks. We tend to think of this as wasteful activity. But here I show that learning about future demand shocks indeed increases their ex ante utility.

PROPOSITION 6. *With access to information about future demand shocks, the ex ante utility of the young generation is improved, while that of the old generation is improved when $K$, $\tau_x$ and $\gamma_{\epsilon x} = \gamma_{\nu x}$ are small enough.*

Learning about the future demand shocks resolves the fallacy of composition. As shown in Figure 4, the ex ante utility improves through the future uncertainty channel. On the aggregate level, less fundamental information acquisition leads to a higher risk premium, while price is more informative and thus investors take less risk in the second period.




**(a)** $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$ **(b)** $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$

**Fig. 4. Information acquisition on future demand shocks improves ex ante expected utility of investors.** Panel (a) displays the contour plot of $\Delta\mathbb{E}[U] = \mathbb{E}[U] - \mathbb{E}[U^B]$ subject to the linear information constraint. Raising information processing capacity or the prior precision of demand shocks enlarges the welfare gain from learning future demand shocks. Panel (b) demonstrates that investor expected utility exhibits weaker concavity with respect to information processing capacity relative to the benchmark model.

### *3.5. Price volatility*

From the perspective of $\mathcal{G}_{t-1}$-investor $i$, the price volatility at the beginning of period $t$ is $\mathbb{V}ar(p_t \mid \hat{\mathcal{I}}^+_{it-1}) = C_t^2/\tau_y + D_t^2/(\tau_x + \tau_{\nu x t-1})$. For an individual asset $j$, its conditional variance

is $\mathbb{V}ar(p_t \mid \hat{\mathcal{I}}^+_{it-1}) = C_t^2/\tau_y + D_t^2/(\tau_x + \tau_{\nu xt-1})$.

Notice that each generation $t'$ of investors only chooses to learn about future demand shocks in period $t'$, which affects market clearing in period $t'+1$. Recall that $\bar{\Sigma}_t^{-1} = [\tau_y + \tau_{\epsilon yt} + \zeta_t^2(\tau_x + \tau_{\epsilon xt})]/r + \tau_y + \hat{\tau}_{\epsilon yt} + \zeta_t^2(\tau_x + \hat{\tau}_{\epsilon xt} + \hat{\tau}_{\nu xt-1})$.

Comparing with the benchmark, due to acquisition of future demand signals at $t-1^+$ by generation $\mathcal{G}_t$, $\tau_{\epsilon xt}$ or $\tau_{\epsilon yt}$ is decreased. Due to acquisition of future demand signals by generation $\mathcal{G}_{t-1}$ at $t-1^+$, $\hat{\tau}_{\epsilon xt}$ or $\hat{\tau}_{\epsilon yt}$ is decreased, but their information choice at $t-2^+$ increases $\hat{\tau}_{\nu xt-1}$.

THEOREM 3 (Price volatility). *The price volatility $\mathbb{V}ar(p_t)$ changes with information processing capacity $K$. Given that $\partial|\zeta_t|/\partial K > 0$,*

*(a) if $\tau_x \geq \bar{\Omega}_x$, then $\partial \mathbb{V}ar(p_t)/\partial K \geq 0$;*

*(b) if $\underline{\Omega}_x < \tau_x < \bar{\Omega}_x$, then*

$$\frac{\partial \mathbb{V}ar(p_t)}{\partial K} = \begin{cases} \geq 0, & \text{if } \zeta_t \leq \underline{\zeta}_t \text{ or } \bar{\underline{\zeta}}_t \leq \zeta_t < 0, \\ < 0, & \text{if } \underline{\zeta}_t < \zeta_t < \bar{\underline{\zeta}}_t; \end{cases}$$

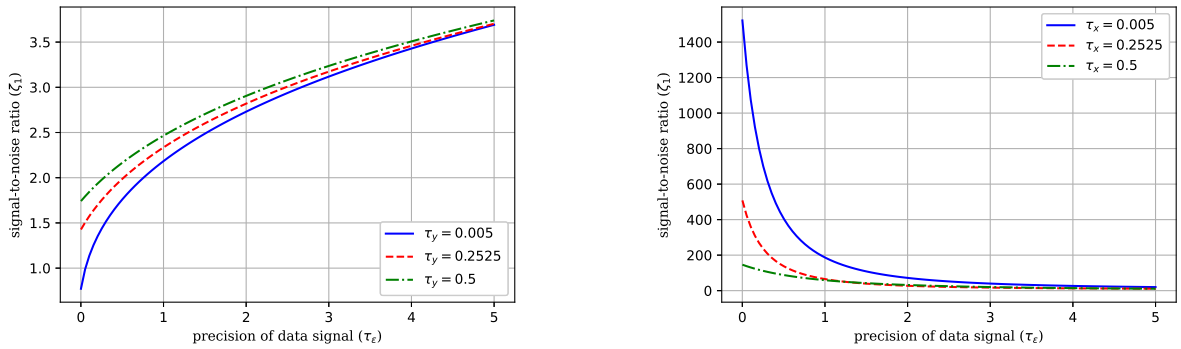
*(c) if $0 < \tau_x \leq \underline{\Omega}_x$, then*

$$\frac{\partial \mathbb{V}ar(p_t)}{\partial K} = \begin{cases} \geq 0, & \text{if } \zeta_t \leq \underline{\zeta}_t, \\ < 0, & \text{if } \underline{\zeta}_t < \zeta_t < 0, \end{cases}$$

*where $\overline{\Omega}_x$, $\underline{\Omega}_x$, $\bar{\underline{\zeta}}_t$, and $\underline{\zeta}_t$ are defined in Appendix C.*

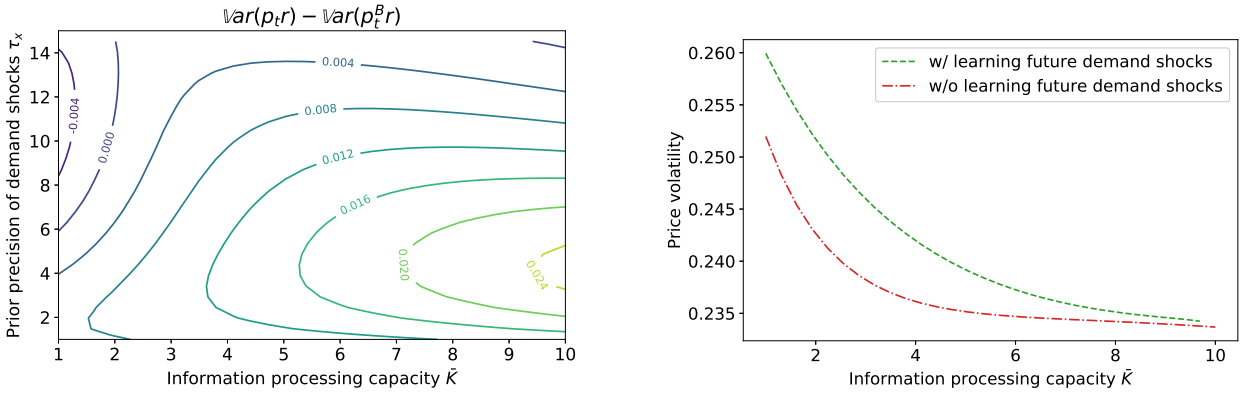
From the perspective of the young generation $\mathcal{G}_t$, price volatility at the beginning of period $t$ is not affected by whether learning about future demand shocks is possible.

Figure 6a shows that for given learning costs $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$, compared with Benchmark I where learning about future demand shocks is impossible, price volatility increases when




**(a)** $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$ **(b)** $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$

**Fig. 5. Access to information about future demand shocks affects price volatility.** Panel (a) reports the contour surface for the volatility wedge $\mathbb{V}ar\left(p_t\, r\right) - \mathbb{V}ar\left(p_t^B\, r\right)$. Panel (b) illustrates how equilibrium price volatility varies with investors' information processing capacity.




**(a)** $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$ **(b)** $\gamma_{\epsilon x} = \gamma_{\nu x} = 0.5$

**Fig. 6. Access to information about future demand shocks affects price volatility.** Panel (a) shows the contour plot of the volatility differential $\mathbb{V}ar\left(p_t\, r\right) - \mathbb{V}ar\left(p_t^B\, r\right)$. Panel (b) traces how equilibrium price volatility varies with information processing capacity.

the price precision of demand shocks $\tau_x$ is low and/or the information processing capacity is high. Figure 6b plots price volatility against the information processing capacity. In each period $t$, there are two generations of investors. Without learning about future demand shocks, the two generations are identical. Learning about future demand shocks introduces more heterogeneity in private information along two margins. On the extensive margin, the older generation will not learn about future demand shocks because they have no subsequent trading period, while the younger generation does. The two generations make different

information choices and receive different private information. On the intensive margin, the older generation has one more signal, so their information set is more diverse. Heterogeneous private information increases price volatility $\mathbb{V}ar(p_t r)$.

## 4. Applicable Scenarios

This model can be applied to a broad range of real-world scenarios where investors are able to learn about future demand shocks. Some examples include portfolio adjustment by index funds upon index recomposition and the BlackRock-BGI acquisition. [10]

*Index recomposition.* As Han et al. (2016) points out, institutional traders, e.g. index funds or ETFs, need to rebalance portfolios for non-informational reasons around index recomposition. The existence of index funds and the requirement that they replicate indices create demand shocks that are non-fundamental to the underlying securities. On the announcement date of index recomposition, investors receive a signal of effective-date demand shocks. Sophisticated investors can even front run the market before the index recomposition announcement by making informed predictions about how index funds will have to rebalance.

*The BGI transaction.* BlackRock, Inc. announced on June 11, 2009 that it agreed to pay $13.5 billion to acquire Barclays Global Investors (BGI), including iShares, from Barclays Bank PLC.[11] When the deal closed on December 1, 2009, BlackRock paid approximately $6.65 billion in cash and issued capital stock valued at $8.53 billion to Barclays.[12]

While Barclays sold BGI to raise funds to strengthen its balance sheet, BlackRock struck the deal with Barclays, according to its Chairman and CEO Laurence Fink, to "broaden the firm's investment capability with passive and quantitative investment strategies,

[10] Zou (2019) used the BlackRock case to provide identification to the documented positive correlation between ETF flows and the price-to-fundamentals relation of underlying stocks.

[11] Barclays PLC Annual Report (2009), `https://home.barclays/content/dam/home-barclays/documents/investor-relations/annualreports/ar2009/2009-form-20-F.pdf` (accessed December 23, 2019).

[12] BlackRock, Inc. Annual Report (2009), `https://ir.blackrock.com/Cache/1001211956.PDF` (accessed December 23, 2019).

particularly exchange-traded funds, and to address client demand for outcome-oriented portfolio strategies and offer institutional and other investors a full spectrum of investment strategies, including passive and quantitative approaches". BGI's products complement BlackRock's fundamental active strategies that employ a bottom-up security selection process led by portfolio managers who use quantitative and fundamental analysis and their market experience to inform their judgment. BlackRock aimed to become a global leader in ETFs. The BGI transaction was expected to increase economies of scale and introduce synergies between alpha and beta investments. Observing the news of the BGI transaction on June 11, sophisticated investors expected the iShares AUM to expand in the following years. For the component stocks of iShares ETFs, these are future demand shocks that are not related to their firm fundamentals.

## 5. Conclusion

In the NREE literature, research has been done to understand how acquisition of information about current demand shocks, current dividend innovations and future dividend innovations affects the market. This article addresses this gap by studying how learning about future demand shocks would affect investors' information and investment choices and their ex ante utility, as well as price informativeness and volatility.

This article shows that with access to signals about future demand shocks, investors choose to learn less about current dividend innovations and demand shocks and allocate precision to learn about future demand shocks. Learning about future demand shocks affects the financial market through the front-running channel and the future uncertainty channel. To isolate the future uncertainty channel, a set of results is derived for the static model where the asset is a series of dividend strips. If future demand shocks are independent of today's demand shocks, then future demand shocks per se do not affect today's asset price. It is *information acquisition* about future demand shocks that makes it relevant to the current price through the *future uncertainty channel*. While the current equilibrium price still does

not aggregate signals about future demand shocks, acquisition of information about future demand shocks affects the risk premium and how the current equilibrium price aggregates signals about current shocks. The future uncertainty channel works through two forces: on the one hand, the aggregate level of dividend information acquisition decreases, which makes dividends riskier and leads to higher required returns; on the other hand, an atomic investor is more certain about the asset payoff in the second period due to the signal about the second-period demand shocks she acquires in the first period. As a result, investors' ex ante expected utility improves. Price volatility increases due to more heterogeneous private information, and learning about future demand shocks makes price more informative in the future but less informative now.

Learning about future demand shocks is important since it breaks the fallacy of composition and improves the investors' ex ante expected utility. It extends the concept of news shocks and has implications for business cycle analysis. It also breaks the conventional fallacy of composition in information economies: wider investor access to private forward information does not necessarily deteriorate aggregate market outcomes. This suggests that predictable demand-side policy signals and information disclosure on future demand shocks can improve welfare.

## Appendix A. Equilibrium Definition

DEFINITION 3. *A rational expectations equilibrium in this economy consists of a sequence of information choices* $\{\tau_{\epsilon yt}, \tau_{\epsilon xt}, \tau_{\nu xt}, \hat{\tau}_{\epsilon yit+1}, \hat{\tau}_{\epsilon xit+1}, \hat{\tau}_{\nu xit+1}\}$*, risky asset prices* $\{p_t\}$*, and portfolio holdings* $\{q_{it}, \hat{q}_{it+1}\}$ *defined for all* $t \in [0, \infty)$*, satisfying the following conditions:*

*(a) At the start of period* $t$*, each* $\mathcal{G}_t$*-investor* $i$ *selects signal precisions* $\tau_{\epsilon yt}$*,* $\tau_{\epsilon xt}$ *and* $\tau_{\nu xt}$ *to maximize ex ante utility* $\mathbb{E}\big[-\exp(-\rho \tilde{W}_{it+1^+}) \;\big|\; \mathcal{I}^+_{t-1}\big]$ *subject to the information capacity constraint* (4)*;*

*(b) In the portfolio choice stage of period* $t$*, each* $\mathcal{G}_t$*-investor* $i$ *allocates* $q_{it}$ *to the risky asset to maximize* $\mathbb{E}\big[-\exp(-\rho \tilde{W}_{it+1^+}) \;\big|\; \mathcal{I}_{it}\big]$*, taking asset prices and realized signals as*

*given, subject to the budget constraint* (3)*;*

*(c) At the beginning of period* $t{+}1$*, each* $\mathcal{G}_t$*-investor* $i$ *chooses signal precisions* $\hat{\tau}_{\epsilon yit+1}$*,* $\hat{\tau}_{\epsilon xit+1}$ *and* $\hat{\tau}_{\nu xit+1}$ *to maximize* $\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it+1^+}) \;\big|\; \hat{\mathcal{I}}_{it}^{+}\big]$*, conditioning on private signals acquired in period* $t$*, subject to the information capacity constraint* (2)*;*

*(d) In the portfolio choice stage of period* $t+1$*, each* $\mathcal{G}_t$*-investor* $i$ *holds* $\hat{q}_{it+1}$ *in the risky asset to maximize* $\mathbb{E}\big[-\exp(-\rho\tilde{W}_{it+1^+}) \;\big|\; \hat{\mathcal{I}}_{it+1}\big]$*, treating current and lagged signal realizations together with asset prices as exogenous, subject to the budget constraint* (1)*;*

*(e) The risky asset price clears the market via the equality of aggregate demand and aggregate supply:*

$$\int_{i\in\mathcal{G}_t} q_{it}\,\mathrm{d}i + \int_{i\in\mathcal{G}_{t-1}} \hat{q}_{it} - q_{it-1}\,\mathrm{d}i = \bar{x}_t + \tilde{x}_t + \int_{i\in\mathcal{G}_{t-2}} \hat{q}_{it-1}\,\mathrm{d}i\,. \tag{A.1}$$

## Appendix B. Model Solutions

### *Appendix B.1. Investors*

This section works out investors' optimal information and portfolio choices, taking price coefficients and market aggregates as given. Without loss of generality, the model is solved in period $t$. Note that although $A_{t+1}$ depends on $\tilde{x}_t$ (B.46), which is not $\mathcal{I}_{t-1}^{+}$- or $\mathcal{I}_{it}$-measurable, the solutions to the optimization problems faced by investors are not affected.

#### *Appendix B.1.1. Bayesian updating*

At the beginning of period $t$, all $\mathcal{G}_t$-investors are identical and thus hold the same belief about period-$t$ payoffs.

Their information choices, $\tau_{\epsilon xt}$, $\tau_{\epsilon yt}$ and $\tau_{\nu xt}$, are also identical (we drop the $i$ subscript for simplicity). Since the signal realizations $\mathcal{G}_t$-investors receive in period $t$ are different, at the $t$-period portfolio choice stage, they hold heterogeneous beliefs about $t$-period payoffs.

Useful conditional variances for simplifying notation are those of the dividend innovation and current and future demand shocks in the current period.

$$V_{yt} \equiv \mathcal{V}(\tilde{d}_t \mid \mathcal{I}_{it}) = 1/\big(\tau_y + \tau_{\epsilon yt} + \tau_{pt}\big), \tag{B.1}$$

$$V_{xt} \equiv \mathcal{V}(\tilde{x}_t \mid \mathcal{I}_{it}) = 1/\big(\tau_x + \tau_{\epsilon xt}\big), \tag{B.2}$$

$$V_{xt}^{\xi} \equiv \mathcal{V}(\tilde{x}_{t+1} \mid \mathcal{I}_{it}) = 1/\big(\tau_x + \tau_{\nu xt}\big). \tag{B.3}$$

At the aggregate level, the period-$(t+1)$ price coefficients are $\mathcal{I}_{it}$-measurable since there is a continuum of investors and thus the price coefficients depend on the distributions of shocks, not the realizations. Therefore, for $\mathcal{G}_t$-investor $i$, at period-$t$ portfolio choice stage, the uncertainty in the period-$t$ excess payoff $\tilde{d}_t + \alpha p_{t+1} - p_t r$ comes from $\tilde{y}_t$, $\tilde{y}_{t+1}$, $\tilde{x}_t$, $\tilde{x}_{t+1}$ and $\tilde{x}_{t+2}$, and the uncertainty in the period-$(t+1)$ excess payoff $\tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1} r$ comes from $\tilde{y}_t$, $\tilde{y}_{t+1}$, $\tilde{y}_{t+2}$, $\tilde{x}_t$, $\tilde{x}_{t+1}$, $\tilde{x}_{t+2}$ and $\tilde{x}_{t+3}$. Thus the corresponding conditional expectation and variance of the period-$t$ excess payoff are

$$\begin{aligned}
\mathcal{V}_t &\equiv \mathbb{V}ar\big(\tilde{d}_t + \alpha p_{t+1} - r p_t \,\big|\, \mathcal{I}_{it}\big) \\
&= \big(1 + \alpha B_{t+1}\big)^2 V_{yt} + \alpha C_{t+1}^2 \tau_y^{-1} + \alpha D_{t+1}^2 V_{xt}^{\xi} + \alpha E_{t+1}^2 \tau_x^{-1},
\end{aligned} \tag{B.4}$$

$$\begin{aligned}
\mathcal{E}_{it} &\equiv \mathbb{E}\big[\tilde{d}_t + \alpha p_{t+1} - r p_t \,\big|\, \mathcal{I}_{it}\big] \\
&= \big(1 + \alpha B_{t+1}\big)\Big[\mu + g d_{t-1} + V_{yt}\big(\tau_{\epsilon yt}\tilde{\eta}_{it}^{y} + \tau_{pt}\tilde{\eta}_{it}^{p}\big)\Big] + \alpha A_{t+1} \\
&\quad + \alpha D_{t+1} V_{xt}^{\xi} \tau_{\nu xt} \tilde{\xi}_{it}^{x} - r p_t.
\end{aligned} \tag{B.5}$$

For $\mathcal{G}_{t-1}$-investors, however, the signal realizations they receive in period $t-1$ are different, which adds another layer of heterogeneity to their information set. For $\mathcal{G}_{t-1}$-investor $i$ at the period-$t$ portfolio choice stage, conditional variances of the current dividend innovation, the

current demand and the future demand shock are

$$\hat{V}_{yit} \equiv \mathbb{V}ar\big(\tilde{d}_t \mid \hat{\mathcal{I}}_{it}\big) = 1/\big(\tau_y + \hat{\tau}_{\epsilon yit} + \hat{\tau}_{pit}\big), \tag{B.6}$$

$$\hat{V}_{xit} \equiv \mathbb{V}ar\big(\tilde{x}_t \mid \hat{\mathcal{I}}_{it}\big) = 1/\big(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\big), \tag{B.7}$$

$$\hat{V}^{\xi}_{xit} \equiv \mathbb{V}ar\big(\tilde{x}_{t+1} \mid \hat{\mathcal{I}}_{it}\big) = 1/\big(\tau_x + \hat{\tau}_{\nu xit}\big), \tag{B.8}$$

For $\mathcal{G}_{t-1}$-investor $i$ at period-$t$ portfolio choice stage, the uncertainty in the period-$t$ excess payoff $\tilde{d}_t + \alpha p_{t+1} - p_t r$ comes from $\tilde{y}_t$, $\tilde{y}_{t+1}$, $\tilde{x}_t$, $\tilde{x}_{t+1}$ and $\tilde{x}_{t+2}$, and the corresponding conditional expectation and variance are

$$\begin{aligned}\hat{\mathcal{V}}_{it} &\equiv \mathbb{V}ar\big(\tilde{d}_t + \alpha p_{t+1} - r p_t \,\big|\, \hat{\mathcal{I}}_{it}\big) \\ &= \big(1 + \alpha B_{t+1}\big)^2 \hat{V}_{yit} + \alpha C^2_{t+1}\tau_y^{-1} + \alpha D^2_{t+1}\hat{V}^{\xi}_{xit} + \alpha E^2_{t+1}\tau_x^{-1},\end{aligned} \tag{B.9}$$

$$\begin{aligned}\hat{\mathcal{E}}_{it} &\equiv \mathbb{E}\big[\tilde{d}_t + \alpha p_{t+1} - r p_t \,\big|\, \hat{\mathcal{I}}_{it}\big] \\ &= \big(1 + \alpha B_{t+1}\big)\Big[\mu + g d_{t-1} + \hat{V}_{yit}\big(\hat{\tau}_{\epsilon yit}\tilde{\hat{\eta}}^{y}_{it} + \hat{\tau}_{pit}\tilde{\hat{\eta}}^{p}_{it}\big)\Big] + \alpha A_{t+1} \\ &\quad + \alpha D_{t+1}\hat{V}^{\xi}_{xit}\hat{\tau}_{\nu xit}\tilde{\hat{\xi}}^{x}_{it} - r p_t.\end{aligned} \tag{B.10}$$

To obtain the expectation and variance of period-$(t+1)$ excess payoffs conditional on $\hat{\mathcal{I}}_{it+1}$, $\hat{V}_{it+1}$ and $\hat{E}_{it+1}$, we only need to change the time subscript $t$ in (B.9) and (B.10) to $t+1$. Notice that $\hat{V}_i$ is $\mathcal{I}^{+}_{t-1}$-measurable, but $\hat{V}_{it+1}$ is only $\mathcal{I}_{it}$-measurable, not $\mathcal{I}^{+}_{t-1}$-measurable. This is left to verify after we have derived the ex-ante expected utility.

*Appendix B.1.2. Optimal portfolio choice*

In this section, I will solve for the optimal portfolio choice of $\mathcal{G}_t$-investor $i$ in both periods $t$ and $t+1$. The solution will apply to any period $t$.

In period $t+1$, $\mathcal{G}_t$-investor $i$ faces the standard one-shot portfolio choice problem, and the solution is

$$\hat{q}_{it+1} = \hat{\mathcal{E}}_{it+1}/(\rho\hat{\mathcal{V}}_{it+1}). \tag{B.11}$$

This solution is taken as given in period $t$. But note that because the uncertainty in $\hat{\mathcal{E}}_{it+1}$ is not fully resolved at the portfolio choice stage of period $t$, the realization of $\hat{q}_{it+1}$ is not known, while its distribution is.

To solve the portfolio choice problem for $\mathcal{G}_t$-investor $i$ in period $t$, substitute (3) into the expected utility $\mathbb{E}\big[-\exp\big(-\rho\tilde{W}_{it+1^+}\big)\,\big|\,\mathcal{I}_{it}\big]$, which is maximized by choosing $q_{it}$. The expected utility can be simplified to

$$\mathbb{E}\big[-\exp\big(-\rho\tilde{W}_{it+1^+}\big)\,\big|\,\mathcal{I}_{it}\big] = -\exp\big(-\rho r^2 W_0\big)\mathbb{E}\Big[\exp\Big(-\rho r q_{it}\big(\tilde{d}_t+\alpha p_{t+1}-rp_t\big)- \rho\hat{q}_{it+1}\big(\tilde{d}_{t+1}+\alpha p_{t+2}-rp_{t+1}\big)\Big)\,\Big|\,\mathcal{I}_{it}\Big].$$

To proceed, we first summarize the properties of the covariance term in the following lemma.

LEMMA 4. *At the $t$-period portfolio choice stage, the covariance of the $t$-period payoff and the $(t+1)$-period payoff of the risky asset for $\mathcal{G}_t$-investor $i$ is*

$$\begin{aligned}\mathcal{C}_t &\equiv \mathbb{C}ov\Big(\hat{\mathcal{E}}_{it+1}\big(\tilde{d}_{t+1}+\alpha p_{t+2}-p_{t+1}r\big),\,\tilde{d}_t+\alpha p_{t+1}-p_t r\,\Big|\,\mathcal{I}_{it}\Big)\\ &= (2\gamma_1\delta_2\gamma_2^2+\delta_1\gamma_2\gamma_2^\dagger)V_{yt}+\big[\gamma_1\delta_3(\gamma_3+\gamma_3^\dagger)+\delta_1\gamma_3\gamma_3^\dagger\big]\tau_y+\\ &\quad\big[\gamma_1\delta_4(\gamma_4+\gamma_4^\dagger)+\delta_1\gamma_4\gamma_4^\dagger\big]V_{xt}^{\xi}+\big[\gamma_1\delta_5(\gamma_5+\gamma_5^\dagger)+\delta_1\gamma_5\gamma_5^\dagger\big]\tau_x. \qquad \text{(B.12)}\end{aligned}$$

*For the static model, $\mathcal{C}_t = 0$.*

*Proof.* For $\mathcal{G}_t$-investor $i$ at her portfolio choice stage, the $t$-period payoff of the risky asset is

$$\tilde{d}_t+\alpha p_{t+1}\mid\mathcal{I}_{it}=\delta_1+\delta_2\tilde{y}_t\mid\tilde{\eta}_{it}^y,\tilde{\eta}_{it}^p+\delta_3\tilde{y}_{t+1}+\delta_4\tilde{x}_{t+1}\mid\tilde{\xi}_{it}^x+\delta_5\tilde{x}_{t+2},$$

where $\delta_1=(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}$, $\delta_2=1+\alpha B_{t+1}$, $\delta_3=\delta_3=\alpha C_{t+1}$, $\delta_4=\alpha D_{t+1}$ and $\delta_5=\alpha E_{t+1}$.

The $(t+1)$-period excess payoff is

$$\tilde{d}_{t+1} + \alpha p_{t+2} - p_{t+1} r \,\big|\, \mathcal{I}_{it} = \gamma_1 + \gamma_2 \tilde{y}_t \,\big|\, \tilde{\eta}^y_{it}, \tilde{\eta}^p_{it} + \gamma_3^\dagger \tilde{y}_{t+1} + \gamma_4^\dagger \tilde{x}_{t+1} \,\big|\, \tilde{\xi}^x_{it} + \gamma_5^\dagger \tilde{x}_{t+1} + \alpha\big(C_{t+2}\tilde{y}_{t+2} + E_{t+2}\tilde{x}_{t+3}\big),$$

where $\gamma_1 = (1+\alpha B_{t+2})(\mu + g\mu + g^2 d_{t-1}) - rB_{t+1}(\mu + d_{t-1}) - rA_{t+1} + \alpha A_{t+2}$, $\gamma_2 = g\,(1+\alpha B_{t+2}) - rB_{t+1}$, $\gamma_3^\dagger = \gamma_3^\dagger = 1 + \alpha B_{t+2} - rC_{t+1}$, $\gamma_4^\dagger = -rD_{t+1}$ and $\gamma_5^\dagger = \alpha D_{t+2} - rE_{t+1}$.

The expected period-$(t+1)$ excess payoff conditional on $\hat{\mathcal{I}}_{it+1}$ is a random variable at the portfolio choice stage of period $t$, denoted by $\hat{\mathcal{E}}_{it+1} \mid \mathcal{I}_{it}$. From (B.10), we have that

$$\hat{\mathcal{E}}_{it+1} \equiv \big(1 + \alpha B_{t+2}\big)\Big[\mu + g\tilde{d}_t + \hat{V}_{yit+1}\big(\hat{\tau}_{eyit+1}\tilde{\hat{\eta}}^y_{it+1} + \hat{\tau}_{pit+1}\tilde{\hat{\eta}}^p_{it+1}\big)\Big] + \alpha A_{t+2} + \alpha D_{t+2}\hat{V}^{\xi}_{xit+1}\hat{\tau}_{\nu xit+1}\tilde{\hat{\xi}}^x_{it+1} - p_{t+1}r.$$

Recall that the period-$t+1$ price signal for $\mathcal{G}_t$-investor $i$ can be written as

$$\begin{aligned}
\tilde{\hat{\eta}}^p_{it+1} &= \tilde{y}_{t+1} + \frac{1}{\zeta_{1t+1}}\Big(\tilde{x}_{t+1} - \mathbb{E}\big[\tilde{x}_{t+1} \,\big|\, \tilde{\eta}^x_{it+1}, \tilde{\hat{\xi}}^x_{it}\big]\Big) \\
&\quad + \frac{1}{\zeta_{2t+1}}\Big(\tilde{x}_{t+2} - \mathbb{E}\big[\tilde{x}_{t+2} \,\big|\, \tilde{\hat{\xi}}^x_{it+1}\big]\Big) \\
&= \tilde{y}_{t+1} + \frac{1}{\zeta_{1t+1}}\Big(\tilde{x}_{t+1} - \hat{V}_{xit+1}\Big(\hat{\tau}_{\epsilon xit+1}\tilde{\eta}^x_{it+1} + \tau_{\nu xt}\tilde{\hat{\xi}}^x_{it}\Big)\Big) \\
&\quad + \frac{1}{\zeta_{2t+1}}\Big(\tilde{x}_{t+2} - \hat{V}^{\xi}_{xit+1}\hat{\tau}_{\nu xit+1}\tilde{\hat{\xi}}^x_{it+1}\Big).
\end{aligned}$$

From the above two equations, we know

$$\hat{\mathcal{E}}_{it+1} \mid \mathcal{I}_{it} = \gamma_1 + \gamma_2 \tilde{y}_t \mid \tilde{\eta}^y_{it}, \tilde{\eta}^p_{it} + \gamma_3 \tilde{y}_{t+1} + \gamma_4 \tilde{x}_{t+1} \mid \tilde{\xi}^x_{it} + \gamma_5 \tilde{x}_{t+2},$$

where

$$\gamma_3 = \big(1+\alpha B_{t+2}\big)\hat{V}_{yit+1}\big(\hat{\tau}_{\epsilon yit+1}+\hat{\tau}_{pit+1}\big) - rC_{t+1},$$
$$\gamma_4 = \alpha D_{t+2}\hat{V}^{\xi}_{xit+1}\hat{\tau}_{\nu xit+1} - rD_{t+1} + \big(1+\alpha B_{t+2}\big)\hat{V}_{yit+1}\hat{\tau}_{pit+1}\hat{V}_{xit+1}\tau_x \,/\, \zeta_{1t+1},$$
$$\gamma_5 = \big(1+\alpha B_{t+2}\big)\hat{V}_{yit+1}\hat{\tau}_{pit+1}\hat{V}^{\xi}_{xit+1}\tau_x \,/\, \zeta_{2t+1} - rE_{t+1}.$$

Using the assumption that $\tilde{y}_t$, $\tilde{y}_{t+1}$, $\tilde{x}_{t+1}$ and $\tilde{x}_{t+2}$ are zero-mean and mutually independent, it follows that all third moments and cross moments are zero. Thus

$$\begin{aligned}\mathcal{C}_t &\equiv \mathbb{C}ov\Big(\hat{\mathcal{E}}_{it+1}\big(\tilde{d}_{t+1}+\alpha p_{t+2}-p_{t+1}r\big),\, \tilde{d}_t+\alpha p_{t+1}-p_t r \,\Big|\, \mathcal{I}_{it}\Big)\\ &= (2\gamma_1\delta_2\gamma_2+\delta_1\gamma_2\gamma_2^{\dagger})V_{yt} + \big[\gamma_1\delta_3(\gamma_3+\gamma_3^{\dagger})+\delta_1\gamma_3\gamma_3^{\dagger}\big]\tau_y +\\ &\quad \big[\gamma_1\delta_4(\gamma_4+\gamma_4^{\dagger})+\delta_1\gamma_4\gamma_4^{\dagger}\big]V^{\xi}_{xt} + \big[\gamma_1\delta_5(\gamma_5+\gamma_5^{\dagger})+\delta_1\gamma_5\gamma_5^{\dagger}\big]\tau_x.\end{aligned}$$

The stationary solution of $B_t$ gives that $\gamma_2 = (1+\alpha B_{t+2})g - rB_{t+1} = 0$. In the static model where $\alpha = 0$, we have $\mathcal{C}_t = 0$. □

On the other hand, the *ex ante* choice of precision of signals in the next period is irrelevant before the realizations of those signals. In other words, $\tau_{\epsilon xit+1}$, $\tau_{\epsilon yit+1}$, $\tau_{\nu xit+1}$ do not enter the objective function and thus do not affect $q_{it}$ and $q_{it+1}$ if investors are still at the portfolio choice stage of period $t$. We summarize it in the following lemma.

Lemma 5. *At the portfolio choice stage of period $t$, $\mathcal{G}_t$-investor $i$ does not need to consider next-period information choices.*

Therefore, the optimization problem can be simplified to

$$\max_{q_{it}} r q_{it}\mathcal{E}_{it} - \frac{\rho}{2}r^2 q_{it}^2\mathcal{V}_t - r\mathcal{C}_t\frac{1}{\hat{\mathcal{V}}_{it+1}}q_{it},$$

which implies

$$q_{it} = \frac{\mathcal{E}_{it}}{\rho r\mathcal{V}_t} - \frac{\mathcal{C}_t}{\rho r\mathcal{V}_t\hat{\mathcal{V}}_{it+1}}. \tag{B.13}$$

In the static case ($\alpha = 0$), $\mathcal{C}_t = 0$, which greatly simplifies the portfolio choices to

$$q_{it} = \frac{\mathbb{E}[\tilde{d}_t \mid \mathcal{I}_{it}] - p_t r}{\rho r \mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it})}, \quad \hat{q}_{it} = \frac{\mathbb{E}[\tilde{d}_t \mid \hat{\mathcal{I}}_{it}] - p_t r}{\rho \mathbb{V}ar(\tilde{d}_t \mid \hat{\mathcal{I}}_{it})}. \tag{B.14}$$

*Appendix B.1.3. Optimal Information Choices*

In this section, I calculate the *ex ante* expected utility of $\mathcal{G}_t$- and $\mathcal{G}_{t-1}$-investors at the beginning of period $t$ when they have not made any period-$t$ information or portfolio choice, and then solve their information choice problems, that is to maximize the *ex ante* expected utility subject to the information capacity constraint. The *ex ante* expected utility of $\mathcal{G}_{t-1}$-investor $i$ is summarized by the following proposition.

PROPOSITION 7. *At the beginning of period $t$, the ex ante utility of $\mathcal{G}_{t-1}$ is*

$$\begin{aligned}\mathbb{E}\big[\exp\big(-\rho r W_{it^+}\big) \,\big|\, \hat{\mathcal{I}}_{it-1^+}\big] = &-\exp\big(-\rho r W_{it-1^+}\big)\big|\hat{\Sigma}_{t-1^+}/\hat{\mathcal{V}}_{it}\big|^{-1/2} \times \\ &\exp\big(-\mu^2_{\hat{\mathcal{E}}_t}/(2\hat{\Sigma}_{t-1^+})\big).\end{aligned} \tag{B.15}$$

*where $\mu_{\hat{\mathcal{E}}_t}$ and $\hat{\Sigma}_{t-1^+}$ are defined by* (B.18) *and* (B.17)*, respectively.*

*Proof.*

Step 1: Calculate $\mathbb{E}\big[-\exp\big(-\rho W_{it^+}\big) \,\big|\, \hat{\mathcal{I}}_{it}\big]$

Conditional on $\hat{\mathcal{I}}_{it}$, the end-of-period-$t$ wealth $W_{it^+}$ is normally distributed, and thus $\mathbb{E}\big[-\exp\big(-\rho W_{it^+}\big) \,\big|\, \hat{\mathcal{I}}_{it}\big] = -\exp\big(-\rho \mathbb{E}\big[W_{it^+} \,\big|\, \hat{\mathcal{I}}_{it}\big] + (\rho^2/2)\mathbb{V}ar\big(W_{it^+} \,\big|\, \hat{\mathcal{I}}_{it}\big)\big)$. Combining the budget constraint $W_{it^+} = rW_{it-1^+} + \hat{q}_{it}\big(\tilde{d}_t + \alpha p_{t+1} - p_t r\big)$ with the portfolio choice of $\mathcal{G}_{t-1}$-investors (8), we have $\mathbb{E}\big[W_{it^+} \,\big|\, \hat{\mathcal{I}}_{it}\big] = rW_{it-1^+} + \hat{\mathcal{E}}^2_{it}/(\rho\hat{\mathcal{V}}_{it})$ and $\mathbb{V}ar\big(W_{it^+} \,\big|\, \hat{\mathcal{I}}_{it}\big) = \hat{\mathcal{E}}^2_{it}/(\rho^2\hat{\mathcal{V}}_{it})$. Therefore,

$$\mathbb{E}\left[-\exp\left(-\rho W_{it^+}\right) \,\middle|\, \hat{\mathcal{I}}_{it}\right] = -\exp\left(-\rho r W_{it-1^+}\right)\exp\left(-\frac{\hat{\mathcal{E}}^2_{it}}{2\hat{\mathcal{V}}_{it}}\right). \tag{B.16}$$

Step 2: Calculate $\mathbb{E}\big[-\exp\big(-\rho W_{it^+}\big) \,\big|\, \hat{\mathcal{I}}_{it-1^+}\big]$

By the law of iterated expectations,

$$\begin{aligned}\mathbb{E}\big[-\exp\big(-\rho W_{it^+}\big)\,\big|\,\hat{\mathcal{I}}_{it-1^+}\big] &= \mathbb{E}\Big[\,\mathbb{E}\big[-\exp\big(-\rho W_{it^+}\big)\,\big|\,\hat{\mathcal{I}}_{it}\big]\,\Big|\,\hat{\mathcal{I}}_{it-1^+}\Big] \\ &= -\exp\big(-\rho r W_{it-1^+}\big)\mathbb{E}\Big[\exp\Big(-\frac{\hat{\mathcal{E}}_{it}^2}{2\hat{\mathcal{V}}_{it}}\Big)\,\Big|\,\hat{\mathcal{I}}_{it-1^+}\Big].\end{aligned}$$

Let $\mu_{\hat{\mathcal{E}}_t} \equiv \mathbb{E}[\hat{\mathcal{E}}_{it} \mid \hat{\mathcal{I}}^+_{it-1}]$ and $\hat{\Sigma}_{t-1^+} \equiv \mathbb{V}ar(\tilde{d}_t + \alpha p_{t+1} - p_t r \mid \hat{\mathcal{I}}^+_{it-1})$. Conditional on $\hat{\mathcal{I}}_{it-1^+}$, $\hat{\mathcal{E}}_{it}$ is still normally distributed. By the law of total variance, $\hat{\Sigma}_{t-1^+} = \mathbb{E}[\hat{\mathcal{V}}_{it} \mid \mathcal{I}^+_{it-1}] + \mathbb{V}ar(\hat{\mathcal{E}}_{it} \mid \mathcal{I}^+_{it-1})$. Since $\hat{V}_{it}$ is $\mathcal{I}^+_{it-1}$-measurable, we have $\mathbb{V}ar(\hat{\mathcal{E}}_{it} \mid \mathcal{I}^+_{it-1}) = \hat{\Sigma}_{t-1^+} - \hat{\mathcal{V}}_{it}$. And the total variance

$$\begin{aligned}\hat{\Sigma}_{t-1^+} &\equiv \mathbb{V}ar\Big(\tilde{d}_t + \alpha p_{t+1} - p_t r \,\Big|\, \hat{\mathcal{I}}_{it-1^+}\Big) \\ &= \big[(1 + \alpha B_{t+1} - rC_t)^2 + \alpha C_{t+1}^2\big]\tau_y + r^2 D_t^2(\tau_x + \tau_{\nu x t-1})^{-1} + \\ &\quad \big[(\alpha D_{t+1} - rE_t)^2 + \alpha E_{t+1}^2\big]\tau_x. \end{aligned} \tag{B.17}$$

To compute the mean of $\hat{\mathcal{E}}_{it}$, by the law of iterated expectations,

$$\begin{aligned}\mu_{\hat{\mathcal{E}}_t} &\equiv \mathbb{E}\big[\hat{\mathcal{E}}_{it}\,\big|\,\hat{\mathcal{I}}_{it-1^+}\big] = \mathbb{E}\big[\tilde{d}_t + \alpha p_{t+1} - p_t r\,\big|\,\hat{\mathcal{I}}_{it-1^+}\big] \\ &= \frac{r}{r - \alpha g}\mu + \alpha A_{t+1} - rA_t - rD_t V^{\xi}_{xt-1}\tau_{\nu x t-1}\xi^x_{it-1}. \end{aligned} \tag{B.18}$$

To work out $\mathbb{E}\big[\exp\big(-\hat{\mathcal{E}}_{it}^2/\big(2\hat{\mathcal{V}}_{it}\big)\big)\,\big|\,\hat{\mathcal{I}}_{it-1^+}\big]$, it is useful to write it in terms of the mean-zero variable $\hat{\mathcal{E}}_{it} - \mu_{\hat{\mathcal{E}}_t}$[13]:

$$\begin{aligned}&\mathbb{E}\left[\exp\left(-\frac{\hat{\mathcal{E}}_{it}^2}{2\hat{\mathcal{V}}_{it}}\right)\,\Big|\,\hat{\mathcal{I}}_{it-1^+}\right] \\ &= \mathbb{E}\left[\exp\left(-\frac{\big(\hat{\mathcal{E}}_{it} - \mu_{\hat{\mathcal{E}}_t}\big)^2}{2\hat{\mathcal{V}}_{it}} - \frac{\big(\hat{\mathcal{E}}_{it} - \mu_{\hat{\mathcal{E}}_t}\big)\mu_{\hat{\mathcal{E}}_t}}{\hat{\mathcal{V}}_{it}} - \frac{\mu^2_{\hat{\mathcal{E}}_t}}{2\hat{\mathcal{V}}_{it}}\right)\,\Big|\,\hat{\mathcal{I}}_{it-1^+}\right].\end{aligned}$$

[13]See Veldkamp (2011). The general formula for the mean of the exponential of a multivariate quadratic form of normal variables is the following: If $z \sim N(0, \Sigma)$, $F$, $G$ and $H$ are non-random, then $\mathbb{E}\big[\exp\big(z'Fz + G'z + H\big)\big] = \big|I - 2\Sigma F\big|^{-1/2}\exp\Big(\big(1/2\big)G'(I - 2\Sigma F)^{-1}\Sigma G + H\Big)$.

Evaluating this expectation yields

$$\mathbb{E}\big[\exp\Big(-\frac{\hat{\mathcal{E}}_{it}^2}{2\hat{\mathcal{V}}_{it}}\Big)\,\big|\,\hat{\mathcal{I}}_{it-1^+}\big] = \left|\frac{\hat{\Sigma}_{t-1^+}}{\hat{\mathcal{V}}_{it}}\right|^{-1/2}\exp\Big(-\frac{\mu_{\hat{\mathcal{E}}_t}^2}{2\hat{\Sigma}_{t-1^+}}\Big).$$

□

The key feature of (B.15) is that period-$t$ information choices only enter through $\hat{\mathcal{V}}_{it}$, which greatly simplifies the optimization problem, as stated in the following lemma.

Lemma 6. *At the beginning of period $t$, the information choice problem facing $\mathcal{G}_{t-1}$-investor $i$ is equivalent to*

$$\begin{aligned}\max_{\{\hat{\tau}_{\epsilon yit},\hat{\tau}_{\epsilon xit},\hat{\tau}_{\nu xit}\}} -\hat{\mathcal{V}}_{it}^{\dagger} \equiv -\alpha D_{t+1}^2\big(\tau_x+\hat{\tau}_{\nu xit}\big)^{-1} \\ -\frac{r^2}{(r-\alpha g)^2}\Bigg[\tau_y+\hat{\tau}_{\epsilon yit}+\Bigg(\frac{1}{\zeta_{1t}^2\big(\tau_x+\hat{\tau}_{\epsilon xit}+\tau_{\nu xt-1}\big)}+ \\ \frac{1}{\zeta_{2t}^2\big(\tau_x+\hat{\tau}_{\nu xit}\big)}\Bigg)^{-1}\Bigg]^{-1}.\end{aligned} \tag{B.19}$$

*subject to the information capacity constraint.*

*Proof.* From (B.18) and (B.17), it is clear that $\mu_{\hat{\mathcal{E}}_t}$ and $\hat{\Sigma}_{t-1^+}$ are functions of model primitives, price coefficients, the precision choice and signal realization for the previous period's future demand signal, but have nothing to do with this period's information choices, $\hat{\tau}_{\epsilon yit}$, $\hat{\tau}_{\epsilon xit}$ and $\hat{\tau}_{\nu xit}$. The information choices enter the *ex ante* utility (B.15) only through $\hat{\mathcal{V}}_{it}$. Therefore, maximizing the *ex ante* utility amounts to minimizing $\hat{\mathcal{V}}_{it}$, or maximizing $-\hat{\mathcal{V}}_{it}$. Using (B.9), and disposing of terms that contain only model primitives and price coefficients, one gets the equivalent objective function $\hat{\mathcal{V}}_{it}^{\dagger}$. □

Proposition 8. *The optimal period-$t$ information choice for $\mathcal{G}_{t-1}$-investor $i$ exists and is unique. Furthermore, it is $\mathcal{I}_{t-1}^{+}$-measurable and thus homogeneous for all $\mathcal{G}_{t-1}$-investors.*

*Proof.* To begin with, let me calculate the marginal utility with respect to $\hat{\tau}_{\epsilon yit}$, $\hat{\tau}_{\epsilon xit}$ and $\hat{\tau}_{\nu xit}$.

$$
\begin{aligned}
\hat{L}_{\epsilon yit} &\equiv -\frac{\partial \hat{\mathcal{V}}_{it}^{\dagger}}{\partial \hat{\tau}_{\epsilon yit}} \\
&= r^2(r-\alpha g)^{-2}\times \\
&\quad \left[\tau_y + \hat{\tau}_{\epsilon yit} + \left[\frac{1}{\zeta_{1t}^2\left(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\right)} + \frac{1}{\zeta_{2t}^2\left(\tau_x + \hat{\tau}_{\nu xit}\right)}\right]^{-1}\right]^{-2}, \\
\hat{L}_{\epsilon xit} &\equiv -\frac{\partial \hat{\mathcal{V}}_{it}^{\dagger}}{\partial \hat{\tau}_{\epsilon xit}} \\
&= r^2(r-\alpha g)^{-2}\zeta_{1t}^{-2}\left(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\right)^{-2}\times \\
&\quad \left[\left[\frac{1}{\zeta_{1t}^2\left(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\right)} + \frac{1}{\zeta_{2t}^2\left(\tau_x + \hat{\tau}_{\nu xit}\right)}\right]\right]^{-2}\times \\
&\quad \left[\tau_y + \hat{\tau}_{\epsilon yit} + \left[\frac{1}{\zeta_{1t}^2\left(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\right)} + \frac{1}{\zeta_{2t}^2\left(\tau_x + \hat{\tau}_{\nu xit}\right)}\right]^{-1}\right]^{-2}, \\
\hat{L}_{\nu xit} &\equiv -\frac{\partial \hat{\mathcal{V}}_{it}^{\dagger}}{\partial \hat{\tau}_{\nu xit}} \\
&= \alpha D_{t+1}^2\left(\tau_x + \hat{\tau}_{\nu xit}\right)^{-2} + r^2(r-\alpha g)^{-2}\zeta_{1t}^{-2}\left(\tau_x + \hat{\tau}_{\nu xit}\right)^{-2}\times \\
&\quad \left[\left[\frac{1}{\zeta_{1t}^2\left(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\right)} + \frac{1}{\zeta_{2t}^2\left(\tau_x + \hat{\tau}_{\nu xit}\right)}\right]\right]^{-2}\times \\
&\quad \left[\tau_y + \hat{\tau}_{\epsilon yit} + \left[\frac{1}{\zeta_{1t}^2\left(\tau_x + \hat{\tau}_{\epsilon xit} + \tau_{\nu xt-1}\right)} + \frac{1}{\zeta_{2t}^2\left(\tau_x + \hat{\tau}_{\nu xit}\right)}\right]^{-1}\right]^{-2}.
\end{aligned}
$$

It is straightforward to verify that $\partial \hat{\mathcal{V}}_{it}^{\dagger}/\partial \hat{\tau}_{\epsilon yit},\ \partial \hat{\mathcal{V}}_{it}^{\dagger}/\partial \hat{\tau}_{\epsilon xit},\ \partial \hat{\mathcal{V}}_{it}^{\dagger}/\partial \hat{\tau}_{\nu xit} > 0$. Computing the second-order partial derivatives further yields $\partial^2 \hat{\mathcal{V}}_{it}^{\dagger}/\partial \hat{\tau}_{\epsilon yit}^2,\ \partial^2 \hat{\mathcal{V}}_{it}^{\dagger}/\partial \hat{\tau}_{\epsilon xit}^2,\ \partial^2 \hat{\mathcal{V}}_{it}^{\dagger}/\partial \hat{\tau}_{\nu xit}^2 < 0$. As a result, $\hat{\mathcal{V}}_{it}^{\dagger}$ and hence the *ex ante* utility function given in Eq.B.15 are strictly increasing and concave with respect to the information choices $\hat{\tau}_{\epsilon yit}$, $\hat{\tau}_{\epsilon xit}$, and $\hat{\tau}_{\nu xit}$. As the information capacity constraint is linear, the solution to the period-$t$ information choice problem for generation-$t$ investors ($\mathcal{G}_{t-1}$) exists and is unique.

The first-order conditions for the constrained optimization problem are

$$\hat{L}_{\epsilon yit} = \lambda \gamma_{\epsilon y}, \tag{B.20}$$

$$\hat{L}_{\epsilon xit} = \lambda \gamma_{\epsilon x}, \tag{B.21}$$

$$\hat{L}_{\nu xit} = \lambda \gamma_{\nu x}, \tag{B.22}$$

$$\gamma_{\epsilon y}\hat{\tau}_{\epsilon yit} + \gamma_{\epsilon x}\hat{\tau}_{\epsilon xit} + \gamma_{\nu x}\hat{\tau}_{\nu xit} = K. \tag{B.23}$$

Dividing (B.20) by (B.21), one can express $\hat{\tau}_{\nu xit}$ in terms of $\hat{\tau}_{\epsilon xit}$, and dividing (B.20) by (B.22), one can express $\hat{\tau}_{\epsilon xit}$ in terms of $\hat{\tau}_{\nu xit}$ and $\hat{\tau}_{\epsilon xit}$. Then plugging them back into (B.23) pins down $\hat{\tau}_{\epsilon xit}$. Because the first-order conditions depend only on model primitives, price coefficients and $\tau_{\nu xt-1}$, and $\tau_{\nu xt-1}$ is by definition $\mathcal{I}^+_{t-1}$-measurable, period-$t$ information choices of $\mathcal{G}_{t-1}$-investor $i$ are also $\mathcal{I}^+_{t-1}$-measurable. □

This means that despite the fact that $\mathcal{I}^+_{it-1}$, the information set $\mathcal{G}_{t-1}$-investors hold at the beginning of period $t$, is heterogeneous, the information relevant to information choices is still homogeneous. Information choices for both periods are determined in the very beginning, and investors within the same generation all make the same information choices for both periods.

LEMMA 7. *The uncertainty in the variance of excess payoff resolves earlier than the realization of corresponding shocks. More specifically,*

*(1)* $\mathcal{V}_t$, $\mathcal{V}_2$ *and* $\mathcal{C}$ *are* $\mathcal{I}^+_{t-1}$*-measurable;*

*(2)* $\hat{\mathcal{V}}_{it+1}$ *is* $\mathcal{I}^+_{t-1}$*-measurable.*

PROPOSITION 9. *The ex ante utility of* $\mathcal{G}_t$ *is*

$$\begin{aligned}
&\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big)\,\big|\,\mathcal{I}_{t-1}^+\big] \\
&= -\left|\frac{\hat{\Sigma}_{t^+}}{\hat{\mathcal{V}}_{it+1}}\right|^{-1/2}\left|1-2\big(\tau_x^{-1}+\tau_{\nu xt}^{-1}\big)F\right|^{-1/2}\times \\
&\quad \exp\left(-\rho r^2 W_{it-1^+}+\frac{5\mathcal{C}_t^2}{8\mathcal{V}_t\hat{\mathcal{V}}_{it+1}^2}-\frac{\mathcal{C}_t^2}{8\Sigma_{t-1^+}^{\xi}\hat{\mathcal{V}}_{it+1}^2}\right)\times \\
&\quad \exp\left(-\frac{1}{2}\frac{G^2}{\big(\tau_x^{-1}+\tau_{\nu xt}^{-1}\big)^{-1}-2F}+H\right),
\end{aligned} \tag{B.24}$$

*where* $F$*,* $G$*, and* $H$ *are defined by* (B.26)*,* (B.27)*, and* (B.28)*, respectively.*

*Proof.* Substituting the optimal portfolio choices into the *ex ante* expected utility, we have

$$\begin{aligned}
&\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right)\,\middle|\,\mathcal{I}_{t-1}^+\right] \\
&= -\exp\left(-\rho r^2 W_0\right)\mathbb{E}\left[\exp\left(-\left(\frac{\mathcal{E}_{it}}{\mathcal{V}_t}-\frac{\mathcal{C}_t}{\mathcal{V}_t\hat{\mathcal{V}}_{it+1}}\right)\left(\tilde{d}_t+\alpha p_{t+1}-p_t r\right)\right.\right. \\
&\qquad \left.\left.-\frac{\hat{\mathcal{E}}_{it+1}}{\hat{\mathcal{V}}_{it+1}}\left(\tilde{d}_{t+1}+\alpha p_{t+2}-p_{t+1}r\right)\right)\right].
\end{aligned}$$

According to Lemma 9, $\hat{\mathcal{V}}_{it+1}$ is $\mathcal{I}_{it}$-measurable. Thus the exponent term in the above expression is a quadratic form of normal variables under $\mathcal{I}_{it}$. Then we derive the *ex ante* expected utility in two steps.

Step 1: Calculate $\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big)\,\big|\,\hat{\mathcal{I}}_{it+1}\big]$

Conditional on $\hat{\mathcal{I}}_{it+1}$, $W_{it+1^+}$ is a normal random variable. Thus

$$\begin{aligned}
&\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right)\,\middle|\,\hat{\mathcal{I}}_{it+1}\right] \\
&= -\exp\left(-\rho\mathbb{E}\big[W_{it+1^+}\,\big|\,\hat{\mathcal{I}}_{it+1}\big]+\frac{\rho^2}{2}\mathbb{V}ar\big(W_{it+1^+}\,\big|\,\hat{\mathcal{I}}_{it+1}\big)\right).
\end{aligned} \tag{B.25}$$

Recall that $W_{it+1^+} = rW_{it^+}+\hat{q}_{it+1}\big(\tilde{d}_{t+1}+\alpha p_{t+2}-p_{t+1}r\big)$. Substituting the optimal portfolio choice, we have $\mathbb{E}\big[W_{it+1^+}\,\big|\,\hat{\mathcal{I}}_{it+1}\big] = rW_{it^+}+\hat{\mathcal{E}}_{it+1}^2/\big(\rho\hat{\mathcal{V}}_{it+1}\big)$ and

$\mathbb{V}ar\big(W_{it+1^+} \,\big|\, \hat{\mathcal{I}}_{it+1}\big) = \hat{\mathcal{E}}^2_{it+1}/\big(\rho^2 \hat{\mathcal{V}}_{it+1}\big)$. Substituting them into (B.25), we have

$$\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right) \,\middle|\, \hat{\mathcal{I}}_{it+1}\right] = -\exp\left(-\rho r W_{it^+}\right) \exp\left(-\frac{\hat{\mathcal{E}}^2_{it+1}}{2\hat{\mathcal{V}}_{it+1}}\right).$$

Step 2: Calculating $\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big) \,\big|\, \mathcal{I}_{it}\big]$.

Applying (7) to period $t+1$, we have

$$\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right) \,\middle|\, \mathcal{I}^+_{it}\right] = -\exp\left(-\rho r W_{it^+}\right) \left|\frac{\hat{\Sigma}_{t^+}}{\hat{\mathcal{V}}_{it+1}}\right|^{-1/2} \exp\left(-\frac{\mu^2_{\hat{\mathcal{E}}_{t+1}}}{2\hat{\Sigma}_{t^+}}\right).$$

On the one hand, note that $\hat{\Sigma}_{t^+}$ is $\mathcal{I}^+_{t-1}$-measurable, and that $\mu_{\hat{\mathcal{E}}_{t+1}}$ is $\mathcal{I}_{it}$-measurable. On the other hand, $W_{it^+} = rW_{it-1^+} + q_{it}\big(\tilde{d}_t + \alpha p_{t+1} - p_t r\big)$, and $q_{it} = \mathcal{E}_{it}/(\rho r \mathcal{V}_t) - \mathcal{C}_t/\big(\rho r \mathcal{V}_t \hat{\mathcal{V}}_{it+1}\big)$. Thus

$$\mathbb{E}\left[\exp\left(-\rho r W_{it^+}\right) \,\middle|\, \mathcal{I}_{it}\right] = \exp\left(-\rho r^2 W_{it-1^+}\right) h_{it},$$

where the rightmost item $h_t$ is the expectation of the exponential of a linear combination of random variables, that is $h_{it} = \exp\Big(-\mathcal{E}^2_{it}/\big(2\mathcal{V}_t\big) + \mathcal{C}_t\mathcal{E}_{it}/\big(2\mathcal{V}_t\hat{\mathcal{V}}_{it+1}\big) + \mathcal{C}^2_t/\big(2\mathcal{V}_t\hat{\mathcal{V}}^2_{it+1}\big)\Big)$. Therefore, by the law of iterated expectations,

$$\begin{aligned}
&\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right) \,\middle|\, \mathcal{I}_{it}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[-\exp\left(-\rho W_{it+1^+}\right) \,\middle|\, \mathcal{I}^+_{it}\right] \,\middle|\, \mathcal{I}_{it}\right] \\
&= -\left|\frac{\hat{\Sigma}_{t^+}}{\hat{\mathcal{V}}_{it+1}}\right|^{-1/2} \exp\left(-\rho r^2 W_{it-1^+} + \frac{\mathcal{C}^2_t}{2\mathcal{V}_t\hat{\mathcal{V}}^2_{it+1}}\right) \times \\
&\quad \exp\left(-\frac{\mathcal{E}^2_{it}}{2\mathcal{V}_t} + \frac{\mathcal{C}_t\mathcal{E}_{it}}{2\mathcal{V}_t\hat{\mathcal{V}}_{it+1}} - \frac{\mu^2_{\hat{\mathcal{E}}_{t+1}}}{2\hat{\Sigma}_{t^+}}\right).
\end{aligned}$$

Step 3: Calculate $\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big) \,\big|\, \mathcal{I}^+_{t-1}, \xi^x_{it}\big]$.

Conditional on $\mathcal{I}^+_{t-1}$ and $\xi^x_t$, randomness in the above equation only comes from $\mathcal{E}_{it}$. Let

$\mu^{\xi}_{\mathcal{E}_t} \equiv \mathbb{E}\left[\mathcal{E}_{it} \mid \mathcal{I}^+_{t-1}, \xi^x_{it}\right]$ and $\Sigma^{\xi}_{t-1^+} \equiv \mathbb{V}ar(\tilde{d}_t + \alpha p_{t+1} - p_t r \mid \mathcal{I}^+_{t-1}, \xi^x_{it})$.

$$\begin{aligned}\mu^{\xi}_{\mathcal{E}_t} &\equiv \mathbb{E}\big[\mathcal{E}_{it} \,\big|\, \mathcal{I}^+_{t-1}, \xi^x_t\big] = \mathbb{E}\big[\tilde{d}_t + \alpha p_{t+1} - p_t r \,\big|\, \mathcal{I}^+_{t-1}, \xi^x_t\big] \\ &= \frac{r}{r-\alpha g}\mu + \alpha A_{t+1} - rA_t + \big(\alpha D_{t+1} - rE_t\big)V^{\xi}_{xt}\tau_{\nu xt}\xi^x_{it}.\end{aligned}$$

And the total variance

$$\begin{aligned}\Sigma^{\xi}_{t-1^+} &\equiv \mathbb{V}ar\big(\tilde{d}_t + \alpha p_{t+1} - p_t r \,\big|\, \hat{\mathcal{I}}_{it-1^+}\big) \\ &= \big(\alpha D_{t+1} - rE_t\big)^2\big(\tau_x + \tau_{\nu xt}\big)^{-1} + \\ &\quad \big[r^2 D_t^2 + \alpha E_{t+1}^2\big]\tau_x + \big[(1 + \alpha B_{t+1} - rC_t)^2 + \alpha C_{t+1}^2\big]\tau_y.\end{aligned}$$

It is useful to write $\mathcal{E}_{it}$ in terms of the mean-zero variable $\mathcal{E}_{it} - \mu^{\xi}_{\mathcal{E}_t}$, that is

$$-\frac{\mathcal{E}_{it}^2}{2\mathcal{V}_t} = -\frac{\big(\mathcal{E}_{it} - \mu^{\xi}_{\mathcal{E}_t}\big)^2}{2\mathcal{V}_t} - \frac{\big(\mathcal{E}_{it} - \mu^{\xi}_{\mathcal{E}_t}\big)\mu^{\xi}_{\mathcal{E}_t}}{\mathcal{V}_t} - \frac{\big(\mu^{\xi}_{\mathcal{E}_t}\big)^2}{2\mathcal{V}_t}.$$

Therefore,

$$\begin{aligned}&\mathbb{E}\big[\exp\big(-\tfrac{\mathcal{E}_{it}^2}{2\mathcal{V}_t} + \tfrac{\mathcal{C}_t\mathcal{E}_{it}}{2\mathcal{V}_t\hat{\mathcal{V}}_{it+1}}\big) \,\big|\, \mathcal{I}^+_{t-1}\big] \\ &= \mathbb{E}\Big[\exp\Big(-\tfrac{\big(\mathcal{E}_{it} - \mu^{\xi}_{\mathcal{E}_t}\big)^2}{2\mathcal{V}_t} + \big(\tfrac{\mathcal{C}_t}{2\mathcal{V}_t\hat{\mathcal{V}}_{it+1}} - \tfrac{\mu^{\xi}_{\mathcal{E}_t}}{\mathcal{V}_t}\big)\big(\mathcal{E}_{it} - \mu^{\xi}_{\mathcal{E}_t}\big) - \tfrac{\big(\mu^{\xi}_{\mathcal{E}_t}\big)^2}{2\mathcal{V}_t} + \tfrac{\mathcal{C}_t\mu^{\xi}_{\mathcal{E}_t}}{2\mathcal{V}_t\hat{\mathcal{V}}_{it+1}}\Big) \,\Big|\, \mathcal{I}^+_{t-1}\Big] \\ &= \exp\Big(\tfrac{\mathcal{C}_t^2}{8\hat{\mathcal{V}}^2_{it+1}}\big(\tfrac{1}{\mathcal{V}_t} - \tfrac{1}{\Sigma^{\xi}_{t-1^+}}\big)\Big)\exp\Big(-\tfrac{\big(\mu^{\xi}_{\mathcal{E}_t}\big)^2}{2\Sigma^{\xi}_{t-1^+}} - \tfrac{\mathcal{C}_t\mu^{\xi}_{\mathcal{E}_t}}{2\Sigma^{\xi}_{t-1^+}\hat{\mathcal{V}}_{it+1}}\Big).\end{aligned}$$

Putting them together, we have

$$\begin{aligned}&\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big) \,\big|\, \mathcal{I}^+_{t-1}, \xi^x_{it}\big] \\ &= \mathbb{E}\Big[\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big) \,\big|\, \mathcal{I}_{it}\big] \,\Big|\, \mathcal{I}^+_{t-1}, \xi^x_{it}\Big] \\ &= -\big|\tfrac{\hat{\Sigma}_{t^+}}{\hat{\mathcal{V}}_{it+1}}\big|^{-1/2}\exp\Big(-\rho r^2 W_{it-1^+} + \tfrac{5\mathcal{C}_t^2}{8\mathcal{V}_t\hat{\mathcal{V}}^2_{it+1}} - \tfrac{\mathcal{C}_t^2}{8\Sigma^{\xi}_{t-1^+}\hat{\mathcal{V}}^2_{it+1}}\Big)\exp(f),\end{aligned}$$

where

$$f = -\frac{\mu^2_{\hat{\mathcal{E}}_{t+1}}}{2\hat{\Sigma}_{t^+}} - \frac{\left(\mu^{\xi}_{\mathcal{E}_t}\right)^2}{2\Sigma^{\xi}_{t-1^+}} - \frac{\mathcal{C}_t \mu^{\xi}_{\mathcal{E}_t}}{2\Sigma^{\xi}_{t-1^+}\hat{\mathcal{V}}_{it+1}}.$$

Step 4: Calculate $\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big)\,\big|\,\mathcal{I}^+_{t-1}\big]$.

Conditional on $\mathcal{I}^+_{t-1}$, the only randomness in $f$ comes from $\mu_{\hat{\mathcal{E}}_{t+11}}$ and $\mu^{\xi}_{\mathcal{E}_t}$, both of which are functions of $\xi^x_{it}$. Let $M = r\mu/(r-\alpha g) + \alpha A_{t+1} - rA_t$. Then

$$f = F^2 \xi^x_{it} + G\xi_{it} + H,$$

where

$$F = -\frac{1}{2}\big(V^{\xi}_{xt}\tau_{\nu xt}\big)^2\Big[\hat{\Omega}_{t^+} r^2 D^2_{t+1} + \Omega^{\xi}_{t-1^+}\big(\alpha D_{t+1} - rE_t\big)^2\Big], \tag{B.26}$$

$$G = V^{\xi}_{xt}\tau_{\nu xt}\Big[\hat{\Omega}_{t^+} M r D_{t+1} - \Omega^{\xi}_{t-1^+}\big(\alpha D_{t+1} - rE_t\big)\Big(M - \frac{\mathcal{C}_t}{2\hat{\mathcal{V}}_{it+1}}\Big)\Big], \tag{B.27}$$

$$H = -\frac{M^2}{2\hat{\Sigma}_{t^+}} - \frac{M^2}{2\Sigma^{\xi}_{t-1^+}} - \frac{\mathcal{C}_t M}{2\Sigma^{\xi}_{t-1}\hat{\mathcal{V}}_{it+1}}. \tag{B.28}$$

The *ex ante* utility for $\mathcal{G}_t$ investors is

$$\begin{aligned}
&\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big)\,\big|\,\mathcal{I}^+_{t-1}\big] \\
&= \mathbb{E}\Big[\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big)\,\big|\,\mathcal{I}^+_{t-1}, \xi^x_{it}\big]\,\Big|\,\mathcal{I}^+_{t-1}\Big] \\
&= -\left|\frac{\hat{\Sigma}_{t^+}}{\hat{\mathcal{V}}_{it+1}}\right|^{-1/2} \big|1 - 2\big(\tau_x^{-1} + \tau_{\nu xt}^{-1}\big)F\big|^{-1/2} \exp\Big(-\frac{1}{2}\frac{G^2}{\big(\tau_x^{-1} + \tau_{\nu xt}^{-1}\big)^{-1} - 2F} + H\Big) \\
&\quad \exp\Big(-\rho r^2 W_{it-1^+} + \frac{5\mathcal{C}_t^2}{8\mathcal{V}_t\hat{\mathcal{V}}^2_{it+1}} - \frac{\mathcal{C}_t^2}{8\Sigma^{\xi}_{t-1^+}\hat{\mathcal{V}}^2_{it+1}}\Big).
\end{aligned}$$

□

One can apply a strictly increasing transformation of the *ex ante* utility:

$$V \overset{\text{def}}{=} -2\ln\Big[\frac{-\mathbb{E}\big[-\exp\big(-\rho W_{it+1^+}\big)\,\big|\,\mathcal{I}^+_{t-1}\big]}{\exp\big(-\rho r^2 W_0\big)}\Big]$$
$$= \ln H_1 + \ln H_2 + H_3 + H_4, \tag{B.29}$$

where

$$H_1 = \frac{\hat{\Sigma}_{t^+}}{\hat{\mathcal{V}}_{it+1}}, \tag{B.30}$$
$$H_2 = 1 + V^{\xi}_{xt}\frac{\tau_{\nu xt}}{\tau_x}\Big[\hat{\Omega}_{t^+}r^2D^2_{t+1} + \Omega^{\xi}_{t-1^+}\big(\alpha D_{t+1} - rE_t\big)^2\Big], \tag{B.31}$$
$$H_3 = -\frac{5\mathcal{C}^2_t}{4\mathcal{V}_t\hat{\mathcal{V}}^2_{it+1}} + \frac{\mathcal{C}^2_t}{4\Sigma^{\xi}_{t-1^+}\hat{\mathcal{V}}^2_{it+1}}, \tag{B.32}$$
$$H_4 = \frac{G^2}{\big(\tau_x^{-1} + \tau_{\nu xt}^{-1}\big)^{-1} - 2F} - H. \tag{B.33}$$

*Appendix B.2. Market Clearing*

In period $t$, $\mathcal{G}_t$-investors make the same information choices, while $\mathcal{G}_{t-1}$-investors choose precision heterogeneously, depending on what they already learned in period $t-1$ about the period-$t$ demand shock. This effectively makes the aggregation of signals and expected payoffs for the two generations different.

When aggregated, the noise in signals cancels out. The price signals of $\mathcal{G}_t$-investors aggregate to

$$\int_{i\in\mathcal{G}_t}\tilde{\eta}^p_{it}\,\mathrm{d}i = \int_{i\in\mathcal{G}_t}\frac{1}{C_t}\Big(p_t - A_t - B_td_{t-1} - D_t\mathbb{E}\big[\tilde{x}_t\,\big|\,\tilde{\eta}^x_{it}\big] - E_t\mathbb{E}\big[\tilde{x}_{t+1}\,\big|\,\tilde{\xi}^x_{it}\big]\Big)\,\mathrm{d}i$$
$$= \frac{1}{2}\Big[\frac{1}{C_t}(p_t - A_t - B_td_{t-1}) - \frac{1}{\zeta_{1t}}V_{xt}\tau_{\epsilon xt}\tilde{x}_t - \frac{1}{\zeta_{2t}}V^{\xi}_{xt}\tau_{\nu xt}\tilde{x}_{t+1}\Big], \tag{B.34}$$

Then the expected period-$t$ excess payoffs for $\mathcal{G}_t$-investors aggregate to

$$\begin{aligned}\int_{i\in\mathcal{G}_t}\mathcal{E}_{it}\,\mathrm{d}i = \frac{1}{2}\Big\{&(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}+(1+\alpha B_{t+1})V_{yt}\tau_{\epsilon yt}\tilde{y}_t+\\ &\alpha D_{t+1}\big(1-V_{xt}^{\xi}\tau_x\big)\tilde{x}_{t+1}-p_t r+(1+\alpha B_{t+1})V_{yt}\tau_{pt}\times\\ &\Big[\frac{1}{C_t}(p_t-A_t-B_t d_{t-1})-\frac{1}{\zeta_{1t}}V_{xt}\tau_{\epsilon xt}\tilde{x}_t-\frac{1}{\zeta_{2t}}V_{xt}^{\xi}\tau_{\nu xt}\tilde{x}_{t+1}\Big]\Big\}. \end{aligned}\tag{B.35}$$

For $\mathcal{G}_{t-1}$-investors, heterogeneity in conditional variances makes things more complicated:

$$\begin{aligned}&\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}\hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}}\tilde{\eta}_{it}^{p}\,\mathrm{d}i\\ =&\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}\hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}}\frac{1}{C_t}\Big(p_t-A_t-B_t d_{t-1}-D_t\mathbb{E}\big[\tilde{x}_t\,\big|\,\tilde{\eta}_{it}^{x},\tilde{\xi}_{it-1}^{x}\big]-\\ &E_t\mathbb{E}\big[\tilde{x}_{t+1}\,\big|\,\tilde{\tilde{\xi}}_{it}^{x}\big]\Big)\,\mathrm{d}i\\ =&\frac{1}{2}\Big[\hat{\omega}_{pt}\frac{1}{C_t}\big(p_t-A_t-B_t d_{t-1}\big)-\frac{1}{\zeta_{1t}}\hat{\mathcal{Z}}_{1t}\tilde{x}_t-\frac{1}{\zeta_{2t}}\hat{\mathcal{Z}}_{2t}\tilde{x}_{t+1}\Big],\end{aligned}\tag{B.36}$$

where

$$\begin{aligned}\hat{\mathcal{Z}}_{1t}&\equiv 2\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}\hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}}\big(1-\hat{V}_{xit}\tau_x\big)\,\mathrm{d}i,\\ \hat{\mathcal{Z}}_{2t}&\equiv 2\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}\hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}}\big(1-\hat{V}_{xit}^{\xi}\tau_x\big)\,\mathrm{d}i,\\ \hat{\omega}_{pt}&\equiv 2\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}\hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i\end{aligned}$$

are known by the market since the distributions of $\hat{\tau}_{\epsilon xit}$ and $\hat{\tau}_{\nu xit}$ are known. Define the average posterior precision of period-$t$ excess payoff to be $\hat{\Omega}_t \equiv 2\int_{i\in\mathcal{G}_{t-1}}\frac{1}{\hat{\mathcal{V}}_i}\,\mathrm{d}i$, the average ratio of the posterior variance of current demand shock to the posterior variance of period-$t$ excess payoff to be $\hat{\chi}_{xt}^{\xi} \equiv 2\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{xit}^{\xi}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i$, and $\hat{\omega}_{yt} \equiv 2\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}\hat{\tau}_{\epsilon yit}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i$. Then the risk

premium of $\mathcal{G}_{t-1}$-investors aggregate to

$$\int_{i\in\mathcal{G}_{t-1}} \frac{\hat{\mathcal{E}}_{it}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i = \frac{1}{2}\Big\{\hat{\Omega}_t\big[(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}-p_t r\big]+ (1+\alpha B_{t+1})\hat{\omega}_{yt}\tilde{y}_t+\alpha D_{t+1}\big(\hat{\Omega}_t-\hat{\chi}_{xt}\tau_x\big)\tilde{x}_{t+1}+ (1+\alpha B_{t+1})\Big[\hat{\omega}_{pt}\frac{1}{C_t}\big[p_t-A_t-B_t d_{t-1}\big]-\frac{1}{\zeta_{1t}}\hat{\mathcal{Z}}_{1t}\tilde{x}_t- \frac{1}{\zeta_{2t}}\hat{\mathcal{Z}}_{2t}\tilde{x}_{t+1}\Big]\Big\}. \tag{B.37}$$

Recall that the market clearing condition is

$$\bar{x}_t+\tilde{x}_t+\int_{i\in\mathcal{G}_{t-2}}\hat{q}_{it-1}\,\mathrm{d}i=\int_{i\in\mathcal{G}_t}q_{it}\,\mathrm{d}i+\int_{i\in\mathcal{G}_{t-1}}(\hat{q}_{it}-q_{it-1})\,\mathrm{d}i\,.$$

Substituting the portfolio choices, multiplying $\rho$ on both sides and moving $\frac{\mathcal{C}_t}{r\mathcal{V}_t\hat{\mathcal{V}}_{it+1}}$ to LHS, we have

$$\begin{aligned}&\rho(\bar{x}_t+\tilde{x}_t)+\int_{i\in\mathcal{G}_t}\frac{\mathcal{C}_t}{r\mathcal{V}_t\hat{\mathcal{V}}_{it+1}}\,\mathrm{d}i+\int_{i\in\mathcal{G}_{t-1}}\left(\frac{\mathcal{E}_{it-1}}{r\mathcal{V}_{t-1}}-\frac{\mathcal{C}_{t-1}}{r\mathcal{V}_{t-1}\hat{\mathcal{V}}_{it}}\right)\mathrm{d}i+\\ &\int_{i\in\mathcal{G}_{t-2}}\frac{\hat{\mathcal{E}}_{it-1}}{\hat{\mathcal{V}}_{it-1}}\,\mathrm{d}i\\ =&\int_{i\in\mathcal{G}_t}\frac{\mathcal{E}_{it}}{r\mathcal{V}_t}\,\mathrm{d}i+\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{\mathcal{E}}_{it}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i\,.\end{aligned}$$

We work out the right-hand side and left-hand side separately. The right-hand side

contains constants, $p_t$, $\tilde{y}_t$, $\tilde{x}_t$ and $\tilde{x}_{t+1}$:

$$\begin{aligned}
\text{RHS} = \frac{1}{2r}\Omega_t\Big\{&(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}+(1+\alpha B_{t+1})V_{yt}\tau_{\epsilon yt}\tilde{y}_t+\\
&\alpha D_{t+1}\big(1-V_{xt}^{\xi}\tau_x\big)\tilde{x}_{t+1}-p_t r+(1+\alpha B_{t+1})V_{yt}\tau_{pt}\times\\
&\Big[\frac{1}{C_t}(p_t-A_t-B_t d_{t-1})-\frac{1}{\zeta_{1t}}V_{xt}\tau_{\epsilon xt}\tilde{x}_t-\frac{1}{\zeta_{2t}}\big(1-V_{xt}^{\xi}\tau_x\big)\tilde{x}_{t+1}\Big]\Big\}+\\
\frac{1}{2}\Big\{&\hat{\Omega}_t\big[(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}-p_t r\big]+\\
&(1+\alpha B_{t+1})\hat{\omega}_{yt}\tilde{y}_t+\alpha D_{t+1}\big(\hat{\Omega}_t-\hat{\chi}_{xt}\tau_x\big)\tilde{x}_{t+1}+\\
&(1+\alpha B_{t+1})\Big[\hat{\omega}_{pt}\frac{1}{C_t}\big[p_t-A_t-B_t d_{t-1}\big]-\frac{1}{\zeta_{1t}}\hat{\mathcal{Z}}_{1t}\tilde{x}_t-\frac{1}{\zeta_{2t}}\hat{\mathcal{Z}}_{2t}\tilde{x}_{t+1}\Big]\Big\}.
\end{aligned}$$

Collecting terms for $\tilde{y}_t$, $\tilde{x}_t$ and $\tilde{x}_{t+1}$, we have

$$\begin{aligned}
\text{RHS} = \frac{1}{2r}\Omega_t\Big\{&(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}-p_t r+\\
&(1+\alpha B_{t+1})V_{yt}\tau_{pt}\Big[\frac{1}{C_t}(p_t-A_t-B_t d_{t-1})\Big]\Big\}+\\
\frac{1}{2}\Big\{&\hat{\Omega}_t\big[(1+\alpha B_{t+1})(\mu+gd_{t-1})+\alpha A_{t+1}-p_t r\big]+\\
&(1+\alpha B_{t+1})\Big[\hat{\omega}_{pt}\frac{1}{C_t}\big[p_t-A_t-B_t d_{t-1}\big]\Big]\Big\}+\\
&(1+\alpha B_{t+1})\bar{\omega}_{yt}\tilde{y}_t-(1+\alpha B_{t+1})\frac{1}{\zeta_{1t}}\bar{\mathcal{Z}}_{1t}\tilde{x}_t+\\
&\Big[\alpha D_{t+1}\big(\bar{\Sigma}_t^{-1}-\bar{\chi}_{xt}\tau_x\big)-(1+\alpha B_{t+1})\frac{1}{\zeta_{2t}}\bar{\mathcal{Z}}_{2t}\Big]\tilde{x}_{t+1}, \qquad \text{(B.38)}
\end{aligned}$$

where

$$\begin{aligned}
\bar{\omega}_{yt} &= \tfrac{1}{2r}\Omega_t V_{yt}\tau_{\epsilon yt}+\tfrac{1}{2}\hat{\omega}_{yt}, & \bar{\chi}_{xt} &= \tfrac{1}{2r}\Omega_t V_{xt}^{\xi}+\tfrac{1}{2}\hat{\chi}_{xt},\\
\bar{\mathcal{Z}}_{2t} &= \tfrac{1}{2r}\Omega_t V_{yt}\tau_{pt}\big(1-V_{xt}^{\xi}\tau_x\big)+\tfrac{1}{2}\hat{\mathcal{Z}}_{2t}, & \bar{\mathcal{Z}}_{1t} &= \tfrac{1}{2r}\Omega_t V_{yt}\tau_{pt}V_{xt}\tau_{\epsilon xt}+\tfrac{1}{2}\hat{\mathcal{Z}}_{1t}.
\end{aligned}$$

The left-hand side contains only constants and $\tilde{x}_t$:

$$\begin{aligned}
\text{LHS} = \rho(\bar{x}_t + \tilde{x}_t) + \frac{1}{2}\hat{\Omega}_{t+1}\frac{\mathcal{C}_t}{r\mathcal{V}_t} - \frac{1}{2}\hat{\Omega}_t\frac{\mathcal{C}_{t-1}}{r\mathcal{V}_{t-1}} + (1+\alpha B_t)\bar{\omega}_{yt-1}\tilde{y}_{t-1} + \\
\frac{1}{2r}\Omega_{t-1}\Big\{(1+\alpha B_t)(\mu + gd_{t-2}) + \alpha A_t - p_{t-1}r + \\
(1+\alpha B_t)V_{yt-1}\tau_{pt-1}\Big[\frac{1}{C_{t-1}}(p_{t-1} - A_{t-1} - B_{t-1}d_{t-2})\Big]\Big\} + \\
\frac{1}{2}\Big\{\hat{\Omega}_{t-1}\big[(1+\alpha B_t)(\mu + gd_{t-2}) + \alpha A_t - p_{t-1}r\big] + \\
(1+\alpha B_t)\Big[\hat{\omega}_{pt-1}\frac{1}{C_{t-1}}\big[p_{t-1} - A_{t-1} - B_{t-1}d_{t-2}\big]\Big]\Big\} - \\
(1+\alpha B_t)\frac{1}{\zeta_{1t-1}}\bar{\mathcal{Z}}_{1t-1}\tilde{x}_{t-1} + \\
\Big[\alpha D_t\big(\bar{\Sigma}_{t-1}^{-1} - \bar{\chi}_{xt-1}\tau_x\big) - (1+\alpha B_t)\frac{1}{\zeta_{2t-1}}\bar{\mathcal{Z}}_{2t-1}\Big]\tilde{x}_t.
\end{aligned}$$

Notice that at the portfolio choice stage of period $t$, $\tilde{y}_{t-1}$ and $\tilde{x}_{t-1}$ are constants. For notational simplicity, let

$$\begin{aligned}
\bar{A}_t = \frac{1}{2}\hat{\Omega}_{t+1}\frac{\mathcal{C}_t}{r\mathcal{V}_t} - \frac{1}{2}\hat{\Omega}_t\frac{\mathcal{C}_{t-1}}{r\mathcal{V}_{t-1}} + \frac{1}{2r}\Omega_{t-1}\Big\{(1+\alpha B_t)(\mu + gd_{t-2}) + \alpha A_t - p_{t-1}r + \\
(1+\alpha B_t)V_{yt-1}\tau_{pt-1}\Big[\frac{1}{C_{t-1}}(p_{t-1} - A_{t-1} - B_{t-1}d_{t-2})\Big]\Big\} + \\
\frac{1}{2}\Big\{\hat{\Omega}_{t-1}\big[(1+\alpha B_t)(\mu + gd_{t-2}) + \alpha A_t - p_{t-1}r\big] + \\
(1+\alpha B_t)\Big[\hat{\omega}_{pt-1}\frac{1}{C_{t-1}}\big[p_{t-1} - A_{t-1} - B_{t-1}d_{t-2}\big]\Big]\Big\} + \\
(1+\alpha B_t)\bar{\omega}_{yt-1}\tilde{y}_{t-1} - (1+\alpha B_t)\frac{1}{\zeta_{1t-1}}\bar{\mathcal{Z}}_{1t-1}\tilde{x}_{t-1}.
\end{aligned}$$

Then

$$\text{LHS} = \bar{A}_t + \rho\bar{x}_t + \rho\tilde{x}_t + \alpha D_t\big[\bar{\Sigma}_{t-1}^{-1} - \bar{\chi}_{xt-1}\tau_x\big]\tilde{x}_t - (1+\alpha B_t)\frac{1}{\zeta_{2t-1}}\bar{\mathcal{Z}}_{2t-1}\tilde{x}_t, \tag{B.39}$$

Then we try to match coefficients of (B.38) and (B.39) with the following linear

equilibrium price relation:

$$p_t = A_t + B_t d_{t-1} + C_t \tilde{y}_t + D_t \tilde{x}_t + E_t \tilde{x}_{t+1}.$$

Define the average market-wide posterior precision as $\bar{\Sigma}_t^{-1} \equiv \frac{1}{2}(\Omega_t/r + \hat{\Omega}_t)$. Move (B.39) to RHS and $\bar{\Sigma}_t^{-1} p_t r$ to LHS, and then multiply by $\bar{\Sigma}_t / r$ on both sides. This yields

$$\begin{aligned}
p_t r = \frac{1}{r}\bar{\Sigma}_t \Big\{ & \frac{1}{2r}\Omega_t \Big[ (1+\alpha B_{t+1})(\mu + g d_{t-1}) + \alpha A_{t+1} + \\
& (1+\alpha B_{t+1}) V_{yt} \tau_{pt} \Big[ \frac{1}{C_t}(p_t - A_t - B_t d_{t-1}) \Big] \Big] + \\
& \frac{1}{2} \Big[ \hat{\Omega}_t \big[ (1+\alpha B_{t+1})(\mu + g d_{t-1}) + \alpha A_{t+1} \big] + \\
& \quad (1+\alpha B_{t+1}) \Big[ \hat{\omega}_{pt} \frac{1}{C_t} \big[ p_t - A_t - B_t d_{t-1} \big] \Big] \Big] + \\
& (1+\alpha B_{t+1}) \bar{\omega}_{yt} \tilde{y}_t - (1+\alpha B_{t+1}) \frac{1}{\zeta_{1t}} \bar{\mathcal{Z}}_{1t} \tilde{x}_t + \\
& \Big[ \alpha D_{t+1} \big( \bar{\Sigma}_t^{-1} - \bar{\chi}_{xt} \tau_x \big) - (1+\alpha B_{t+1}) \frac{1}{\zeta_{2t}} \bar{\mathcal{Z}}_{2t} \Big] \tilde{x}_{t+1} - \\
& \bar{A} - \rho \tilde{x}_t - \alpha D_t \big[ \bar{\Sigma}_{t-1}^{-1} - \bar{\chi}_{xt-1} \tau_x \big] \tilde{x}_t + (1+\alpha B_t) \frac{1}{\zeta_{2t-1}} \bar{\mathcal{Z}}_{2t-1} \tilde{x}_t \Big\}.
\end{aligned}$$

The coefficients on $p_t$ and $B_t d_{t-1}$ on the right-hand side differ only in sign. Define $\bar{\omega}_{pt} = \frac{1}{2r}\Omega_t V_{yt} \tau_{pt} + \frac{1}{2}\hat{\omega}_{pt}$. Again, move the $p_t$ term to LHS and scale down so that $p_t r$ on LHS has coefficient 1:

$$\begin{aligned}
p_t r = \Big( 1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1}) \bar{\omega}_{pt} \frac{1}{C_t} \Big)^{-1} \times \frac{1}{r}\bar{\Sigma}_t \Big\{ & \\
& \bar{\Sigma}_t^{-1} \big[ (1+\alpha B_{t+1})(\mu + g d_{t-1}) + \alpha A_{t+1} \big] + \\
& (1+\alpha B_{t+1}) \bar{\omega}_{pt} \frac{1}{C_t}(-A_t - B_t d_{t-1}) + \\
& (1+\alpha B_{t+1}) \bar{\omega}_{yt} \tilde{y}_t - (1+\alpha B_{t+1}) \frac{1}{\zeta_{1t}} \bar{\mathcal{Z}}_{1t} \tilde{x}_t + \\
& \Big[ \alpha D_{t+1} \big( \bar{\Sigma}_t^{-1} - \bar{\chi}_{xt}^{\xi} \tau_x \big) - (1+\alpha B_{t+1}) \frac{1}{\zeta_{2t}} \bar{\mathcal{Z}}_{2t} \Big] \tilde{x}_{t+1} - \\
& \bar{A} - \rho \tilde{x}_t - \alpha D_t \big[ \bar{\Sigma}_{t-1}^{-1} - \bar{\chi}_{xt-1}^{\xi} \tau_x \big] \tilde{x}_t + (1+\alpha B_t) \frac{1}{\zeta_{2t-1}} \bar{\mathcal{Z}}_{2t-1} \tilde{x}_t \Big\}. \qquad \text{(B.40)}
\end{aligned}$$

***$B_t$***:

From (B.40), it is easy to identify $B_t$, the coefficient on $d_{t-1}$:

$$B_t = \frac{1}{r}\bar{\Sigma}_t\Big(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big)^{-1}\times \\ \Big[\bar{\Sigma}_t^{-1}(1+\alpha B_{t+1})g - (1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}B_t\Big].$$

Moving all $B_t$ terms to the left-hand side, we have

$$B_t = \Big[1 + \Big(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big)^{-1}\frac{1}{r}\bar{\Sigma}_t(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big]^{-1}\times \\ \Big(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big)^{-1}\frac{1}{r}g(1+\alpha B_{t+1}).$$

Since

$$\Big[1 + \Big(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big)^{-1}\frac{1}{r}\bar{\Sigma}_t(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big]^{-1} \\ \times \Big(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big)^{-1} = 1,$$

it simplifies to

$$B_t = \frac{1}{r}g(1+\alpha B_{t+1}). \tag{B.41}$$

This is a linear difference equation and has a stationary solution[14]

$$B = \frac{g}{r - \alpha g}.$$

***$A_t$***:

[14]The stationary solution is obtained by setting $B_t = B$, $\forall t$ in (B.41) and solving for $B$.

From (B.40), the constant term in the equilibrium price relation is

$$\begin{aligned} A_t = &\left(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\right)^{-1} \\ &\times \frac{1}{r}\bar{\Sigma}_t\left[\bar{\Sigma}_t^{-1}(1+\alpha B_{t+1})\mu + \alpha A_{t+1} - (1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}A_t - \bar{A}_t\right]. \end{aligned}$$

Moving all $A_t$ terms to the left-hand side, we have

$$A_t = \frac{1}{r}\left[(1+\alpha B_{t+1})\mu - \rho\bar{x}_t - \bar{A}_t(A_{t+1}, A_{t+2}) + \alpha A_{t+1}\bar{\Sigma}_t\right]. \tag{B.42}$$

To emphasize the dependency of $\bar{A}_t$ on $A_{t+1}$ and $A_{t+2}$, I write $\bar{A}_t(A_{t+1}, A_{t+2})$ instead of $\bar{A}_t$. Plugging in the stationary solution of $B_t$, we have

$$A_t = \frac{\mu}{r-\alpha g} - \frac{\rho\bar{x}_t}{r} - \frac{1}{r}\bar{A}_t(A_{t+1}, A_{t+2}) + \frac{\bar{\Sigma}_t}{r}\alpha A_{t+1}\bar{\Sigma}_t.$$

**$\underline{C_t}$**:

The coefficient on $\tilde{y}_t$ in (B.40) is

$$C_t = \left(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\right)^{-1}\frac{1}{r}\bar{\Sigma}_t(1+\alpha B_{t+1})\bar{\omega}_{yt}.$$

Multiplying $\left(1 - \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\right)$ on both sides and rearranging, we have

$$C_t = \frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\left(\bar{\omega}_{yt} + \bar{\omega}_{pt}\right).$$

Since

$$\begin{aligned} V_{yt}(\tau_{\epsilon yt} + \tau_{pt}) &= 1 - V_{yt}\tau_y, \\ \hat{\omega}_{yt} + \hat{\omega}_{pt} &= 2\int_{i\in\mathcal{G}_{t-1}} \frac{\hat{V}_{yit}(\hat{\tau}_{\epsilon yit} + \hat{\tau}_{pit})}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i = \hat{\Omega}_t - 2\tau_y\int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i, \end{aligned}$$

we have

$$C_t = \frac{1}{r-\alpha g}\left(1-\bar{\Sigma}_t\bar{\chi}_{yt}\tau_y\right),$$

where

$$\bar{\chi}_{yt} \equiv \frac{1}{r}\Omega_t\frac{V_{yt}}{2} + \int_{i\in\mathcal{G}_{t-1}}\frac{\hat{V}_{yit}}{\hat{\mathcal{V}}_{it}}\,\mathrm{d}i\,.$$

***$D_t$*:**

The coefficient on $\tilde{x}_t$ in the price relation is

$$D_t = \frac{1}{r}\bar{\Sigma}_t\Big(1-\frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\Big)^{-1}\times$$
$$\Big[-\rho-\alpha D_t\big(\bar{\Sigma}_{t-1}^{-1}-\bar{\chi}_{xt-1}^{\xi}\tau_x\big)-(1+\alpha B_{t+1})\frac{1}{\zeta_{1t}}\bar{\mathcal{Z}}_{1t}+(1+\alpha B_t)\frac{1}{\zeta_{2t-1}}\bar{\mathcal{Z}}_{2t-1}\Big].$$

Multiplying $\left(1-\frac{\bar{\Sigma}_t}{r}(1+\alpha B_{t+1})\bar{\omega}_{pt}\frac{1}{C_t}\right)$ on both sides, we have

$$D_t = \frac{\bar{\Sigma}_t}{r-\alpha g}\frac{1}{\zeta_{1t}}\left(\bar{\omega}_{pt}-\bar{\mathcal{Z}}_{1t}\right)-\frac{\rho\bar{\Sigma}_t}{r}+\frac{\bar{\Sigma}_t}{r-\alpha g}\frac{1}{\zeta_{2t-1}}\bar{\mathcal{Z}}_{2t-1}+$$
$$\alpha D_t\frac{\bar{\Sigma}_t}{r}\left(\bar{\Sigma}_{t-1}^{-1}-\bar{\chi}_{xt-1}^{\xi}\tau_x\right).$$

We can then solve for $D_t$ straightforwardly:

$$D_t = \left[1-\alpha\frac{\bar{\Sigma}_t}{r}\left(\bar{\Sigma}_{t-1}^{-1}-\bar{\chi}_{xt-1}^{\xi}\tau_x\right)\right]^{-1}\times$$
$$\left[\frac{\bar{\Sigma}_t}{r-\alpha g}\frac{1}{\zeta_{1t}}\left(\bar{\omega}_{pt}-\bar{\mathcal{Z}}_{1t}\right)-\frac{\rho\bar{\Sigma}_t}{r}+\frac{\bar{\Sigma}_t}{r-\alpha g}\frac{1}{\zeta_{2t-1}}\bar{\mathcal{Z}}_{2t-1}\right].$$

For a better understanding of $\bar{\omega}_{pt}-\bar{\mathcal{Z}}_{1t}$, let $\chi_{yt}\equiv\Omega_t V_{yt}$, which characterizes the proportion of excess payoff uncertainty that can be explained by the uncertainty that comes from dividend innovation. When $\alpha=0$, the risky asset is short-lived and all the uncertainty comes from the dividend innovation. Thus in this case $\chi_{yt}=1$. Notice that $\tau_{pt}=\frac{1}{\zeta_{1t}^{-2}V_{xt}+\zeta_{2t}^{-2}V_{xt}^{\xi}}=\frac{\zeta_{1t}^2}{V_{xt}}-\frac{\zeta_{1t}^4/V_{xt}^2}{\zeta_{1t}^2/V_{xt}+\zeta_{2t}^2/V_{xt}^{\xi}}$. Thus $\zeta_{1t}^{\dagger}\equiv\frac{\tau_{pt}V_{xt}}{\zeta_{1t}}=\zeta_{1t}-\frac{\zeta_{1t}^3/V_{xt}}{\zeta_{1t}^2/V_{xt}+\zeta_{2t}^2/V_{xt}^{\xi}}$ is the future-uncertainty-adjusted signal-to-current-noise ratio. Then define the

market-wide average future-uncertainty-adjusted dividend-to-total-uncertainty-weighted signal-to-current-noise ratio to be

$$\bar{\zeta}_{1t}^{\dagger} \equiv \frac{1}{\zeta_{1t}} \left[ \frac{1}{2r} \Omega_t V_{yt} \tau_{pt} V_{xt} + \int_{i \in \mathcal{G}_{t-1}} \frac{\hat{V}_{yit} \hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}} \hat{V}_{xit} \, \mathrm{d}i \right]. \tag{B.43}$$

When $\alpha = 0$, $\bar{\zeta}_{1t}^{\dagger} = \frac{r+1}{2r} \zeta_t$. Then $(\bar{\omega}_{pt} - \bar{\mathcal{Z}}_{1t})/\zeta_{1t} = \bar{\zeta}_{1t}^{\dagger} \tau_x$.

***$E_t$***:

The coefficient on $\tilde{x}_{t+1}$ in the price relation is

$$E_t = \frac{1}{r} \bar{\Sigma}_t \left( 1 - \frac{\bar{\Sigma}_t}{r} (1 + \alpha B_{t+1}) \bar{\omega}_{pt} \frac{1}{C_t} \right)^{-1} \times \left[ \alpha D_{t+1} \left( \bar{\Sigma}_t^{-1} - \bar{\chi}_{xt}^{\xi} \tau_x \right) - (1 + \alpha B_{t+1}) \frac{1}{\zeta_{2t}} \bar{\mathcal{Z}}_{2t} \right].$$

Multiplying both sides by $\left( 1 - \frac{\bar{\Sigma}_t}{r} (1 + \alpha B_{t+1}) \bar{\omega}_{pt} \frac{1}{C_t} \right)$ and rearranging, we have

$$E_t = \frac{\bar{\Sigma}_t}{r - \alpha g} \frac{1}{\zeta_{2t}} \left( \bar{\omega}_{pt} - \bar{\mathcal{Z}}_{2t} \right) + \frac{1}{r} \alpha D_{t+1} \left( 1 - \bar{\Sigma}_t \bar{\chi}_{xt}^{\xi} \tau_x \right).$$

Notice that $\tau_{pt} = \frac{1}{\zeta_{1t}^{-2} V_{xt} + \zeta_{2t}^{-2} V_{xt}^{\xi}} = \frac{\zeta_{2t}^2}{V_{xt}^{\xi}} - \frac{\zeta_{2t}^4 / \left( V_{xt}^{\xi} \right)^2}{\zeta_{1t}^2 / V_{xt} + \zeta_{2t}^2 / V_{xt}^{\xi}}$. Thus $\zeta_{2t}^{\dagger} \equiv \frac{\tau_{pt} V_{xt}^{\xi}}{\zeta_{2t}} = \zeta_{2t} - \frac{\zeta_{2t}^3 / V_{xt}^{\xi}}{\zeta_{1t}^2 / V_{xt} + \zeta_{2t}^2 / V_{xt}^{\xi}}$ is the future-uncertainty-adjusted signal-to-current-noise ratio. Then define the market-wide average future-uncertainty-adjusted dividend-to-total-uncertainty-weighted signal-to-current-noise ratio as

$$\bar{\zeta}_{2t}^{\dagger} \equiv \frac{1}{\zeta_{2t}} \left[ \frac{1}{2r} \Omega_t V_{yt} \tau_{pt} V_{xt}^{\xi} + \int_{i \in \mathcal{G}_{t-1}} \frac{\hat{V}_{yit} \hat{\tau}_{pit}}{\hat{\mathcal{V}}_{it}} \hat{V}_{xit}^{\xi} \, \mathrm{d}i \right]. \tag{B.44}$$

We can then write $E_t$ as

$$E_t = \frac{1}{r} \alpha D_{t+1} \left( 1 - \bar{\Sigma}_t \bar{\chi}_{xt}^{\xi} \tau_x \right) + \frac{\bar{\Sigma}_t}{r - \alpha g} \bar{\zeta}_{2t}^{\dagger} \tau_x. \tag{B.45}$$

In the static case, we have $E_t = \bar{\Sigma}_t \frac{E_t}{C_t} \tau_x$, which implies that $E_t = 0$.

*Appendix B.3. Existence of Linear Equilibrium*

The price coefficients are

$$A_t = \frac{\mu}{r-\alpha g} - \frac{\rho \bar{x}_t}{r} - \frac{1}{r}\bar{A}_t(A_{t+1}, A_{t+2}) + \frac{\bar{\Sigma}_t}{r}\alpha A_{t+1}\bar{\Sigma}_t, \tag{B.46}$$

$$B = \frac{g}{r-\alpha g}, \tag{B.47}$$

$$C_t = \frac{1}{r-\alpha g}\left(1 - \bar{\Sigma}_t \bar{\chi}_{yt}\tau_y\right), \tag{B.48}$$

$$D_t = \left[1 - \alpha\frac{\bar{\Sigma}_t}{r}\left(\bar{\Sigma}_{t-1}^{-1} - \bar{\chi}_{xt-1}^{\xi}\tau_x\right)\right]^{-1}\left[\frac{\bar{\Sigma}_t}{r-\alpha g}\bar{\zeta}_{1t}^{\dagger}\tau_x - \frac{\rho\bar{\Sigma}_t}{r} + \frac{\bar{\Sigma}_t}{r-\alpha g}\frac{1}{\zeta_{2t-1}}\bar{\mathcal{Z}}_{2t-1}\right], \tag{B.49}$$

$$E_t = \frac{1}{r}\alpha D_{t+1}\left(1 - \bar{\Sigma}_t\bar{\chi}_{xt}^{\xi}\tau_x\right) + \frac{\bar{\Sigma}_t}{r-\alpha g}\bar{\zeta}_{2t}^{\dagger}\tau_x. \tag{B.50}$$

## Appendix C. Proof of Results

Proposition 10. *In the benchmark model, under Assumption 1:*

*(a) An equilibrium exists if* $K \leq \frac{\rho r}{r+1}\sqrt{\gamma_{\epsilon y}\gamma_{\epsilon x}}$.

*(b) In both periods, investors make identical information choices* $\tau_{\epsilon yt}^B = \tau_{\epsilon y,it+1}^B = K/\gamma_{\epsilon y}$ *and* $\tau_{\epsilon xt}^B = \tau_{\epsilon x,it+1}^B = 0$ *if* $K < \rho r\sqrt{\gamma_{\epsilon y}\gamma_{\epsilon x}}/(r+1)$, *and arbitrarily allocate precision to* $\tau_{\epsilon yt}^B$ *and* $\tau_{\epsilon xt}^B$ *if* $K = \rho r\sqrt{\gamma_{\epsilon y}\gamma_{\epsilon x}}/(r+1)$.

*(c) The signal-to-noise ratio* $\zeta_t^B = -K(r+1)/(\rho\, r\, \gamma_{\epsilon y})$.

*(d) The posterior average variance satisfies* $\left(\bar{\Sigma}_t^B\right)^{-1} = (r+1)\left[\tau_y + \left(\zeta_t^B\right)^2\tau_x\right]/r - \rho\zeta_t^B$.

*Proof of Proposition 10.* In the benchmark case,

$$\frac{r+1}{r}\tau_{\epsilon xt}\zeta_t^2 + \rho\zeta_t + \frac{r+1}{r}\tau_{\epsilon yt} = 0.$$

Plugging in the solution for $\tau_{\epsilon xt}^B$ and $\tau_{\epsilon yt}^B$, we have

$$\rho\zeta_t = -\frac{r+1}{r}\sqrt{\frac{K}{\gamma_{\epsilon y}}\left[1 + \frac{\gamma_{\epsilon y}}{\gamma_{\epsilon x}}\zeta_t^4\right]} < 0, \tag{C.1}$$

or

$$\left(\frac{r+1}{r}\right)^2 \frac{K}{\gamma_{\epsilon x}}\zeta_t^4 - \rho^2\zeta_t^2 + \left(\frac{r+1}{r}\right)^2 \frac{K}{\gamma_{\epsilon y}} = 0.$$

This is a quadratic equation of $\zeta_t^2$. We pick the solution which is consistent with the limit. Thus

$$\zeta_t^B = -\sqrt{\frac{\rho^2 - \sqrt{\rho^4 - 4\frac{\gamma_{\epsilon y}}{\gamma_{\epsilon x}}\left(\frac{K}{\gamma_{\epsilon y}}\right)^2\left(\frac{r+1}{r}\right)^4}}{2\frac{\gamma_{\epsilon y}}{\gamma_{\epsilon x}}\frac{K}{\gamma_{\epsilon y}}\left(\frac{r+1}{r}\right)^2}}.$$

□

PROPOSITION 11. *In the benchmark model, under Assumption 1:*

*(a) Price coefficients $C_t^B > 0$, $D_t^B < 0$.*

*(b) The equilibrium signal-to-noise ratio $|\zeta_t|$ rises with information capacity $K$, falls with demand-shock learning cost $\gamma_{\epsilon x}$, and is invariant to prior precisions $\tau_y$ and $\tau_x$.*

*(c) Posterior average precision $(\bar{\Sigma}_t^B)^{-1}$ increases in $K$, the prior precision of the dividend innovation $\tau_y$, the prior precision of the demand shock $\tau_x$, and decreases in demand learning cost $\gamma_{\epsilon x}$.*

*(d) A larger $K$ raises precision assigned to both signals, $\tau_{\epsilon xt}^B$ and $\tau_{\epsilon yt}^B$. A higher $\gamma_{\epsilon x}$ increases the precision of the current dividend innovation $\tau_{\epsilon yt}^B$ and reduces the precision of the current demand shock $\tau_{\epsilon xt}^B$. Information allocations do not respond to prior uncertainty levels.*

*Proof of Proposition 11.* Recall that $\left(\bar{\Sigma}_t^B\right)^{-1} = \frac{r+1}{r}\left[\tau_y + \left(\zeta_t^B\right)^2\tau_x\right] - \rho\zeta_t^B$. Taking the partial derivative with respect to $|\zeta_t^B|$, we have

$$\frac{\partial\left(\bar{\Sigma}_t^B\right)^{-1}}{\partial|\zeta_t^B|} = 2\frac{r+1}{r}|\zeta_t^B|\tau_x + \rho > 0.$$

By the chain rule, we have

$$\begin{aligned}
\frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial K} &= \frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial |\zeta_t|}\frac{\partial |\zeta_t^B|}{\partial K} > 0, \\
\frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial \gamma_{\epsilon x}} &= \frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial |\zeta_t|}\frac{\partial |\zeta_t^B|}{\partial \gamma_{\epsilon x}} < 0, \\
\frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial \tau_x} &= \frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial |\zeta_t|}\frac{\partial |\zeta_t^B|}{\partial \tau_x} + \frac{r+1}{r}\left(\zeta_t^B\right)^2 > 0.
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{V}(p_t)}{\partial K} = 2\Bigg[ & \frac{C_t^B}{\tau_y}\frac{\partial C_t^B}{\partial \bar{\Sigma}_t^B}\frac{\partial \bar{\Sigma}_t^B}{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}\frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial |\zeta_t|} \\
& + \frac{D_t^B}{\tau_x}\left(\frac{\partial D_t^B}{\partial \bar{\Sigma}_t^B}\frac{\partial \bar{\Sigma}_t^B}{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}\frac{\partial \left(\bar{\Sigma}_t^B\right)^{-1}}{\partial |\zeta_t^B|} + \frac{\partial D_t^B}{\partial |\zeta_t^B|}\right)\Bigg]\frac{\partial |\zeta_t^B|}{\partial K}.
\end{aligned}$$

Same as in the main model.

$$\frac{\partial \tau_{\epsilon y t}^B}{\partial K} = \frac{\partial \tau_{\epsilon y t}^B}{\partial K} + \frac{\partial \tau_{\epsilon y t}^B}{\partial \zeta_t^B}\frac{\partial \zeta_t^B}{\partial K}.$$

The first term

$$\frac{\partial \tau_{\epsilon y t}^B}{\partial K} = \frac{1}{2\gamma_{\epsilon y}\left(\frac{\gamma_{\epsilon y}\zeta^4}{\gamma_{\epsilon x}}+1\right)\sqrt{\frac{K}{\gamma_{\epsilon y}\left(\frac{\gamma_{\epsilon y}\zeta^4}{\gamma_{\epsilon x}}+1\right)}}} > 0.$$

$$\frac{\mathrm{d}\tau^B_{\epsilon xt}}{\mathrm{d}K} = \underbrace{\frac{\partial \tau^B_{\epsilon xt}}{\partial K}}_{>0} + \underbrace{\frac{\partial \tau^B_{\epsilon xt}}{\partial \zeta^B_t}}_{<0}\underbrace{\frac{\partial \zeta^B_t}{\partial K}}_{<0} > 0,$$
$$\frac{\mathrm{d}\tau^B_{\epsilon yt}}{\mathrm{d}K} = \underbrace{\frac{\partial \tau^B_{\epsilon yt}}{\partial K}}_{>0} + \underbrace{\frac{\partial \tau^B_{\epsilon yt}}{\partial \zeta^B_t}}_{>0}\underbrace{\frac{\partial \zeta^B_t}{\partial K}}_{<0},$$
$$\frac{\mathrm{d}\tau^B_{\epsilon xt}}{\mathrm{d}\gamma_{\epsilon x}} = \underbrace{\frac{\partial \tau^B_{\epsilon xt}}{\partial \gamma_{\epsilon x}}}_{>0} + \underbrace{\frac{\partial \tau^B_{\epsilon xt}}{\partial \zeta^B_t}}_{<0}\underbrace{\frac{\partial \zeta^B_t}{\partial \gamma_{\epsilon x}}}_{>0},$$
$$\frac{\mathrm{d}\tau^B_{\epsilon yt}}{\mathrm{d}\gamma_{\epsilon x}} = \underbrace{\frac{\partial \tau^B_{\epsilon yt}}{\partial \gamma_{\epsilon x}}}_{<0} + \underbrace{\frac{\partial \tau^B_{\epsilon yt}}{\partial \zeta^B_t}}_{>0}\underbrace{\frac{\partial \zeta^B_t}{\partial \gamma_{\epsilon x}}}_{>0};$$

$$\frac{\partial \zeta^B_t}{\partial \gamma_{\epsilon x}} = \frac{A - B}{2\sqrt{2}C},$$

where

$$A = \frac{\rho^2 - \sqrt{\rho^4 - \dfrac{4\left(\frac{r+1}{r}\right)^4 K^2}{\gamma_{\epsilon x}\gamma_{\epsilon y}}}}{\left(\frac{r+1}{r}\right)^2 K},$$
$$B = \frac{2\left(\frac{r+1}{r}\right)^2 K}{\gamma_{\epsilon x}\gamma_{\epsilon y}\sqrt{\rho^4 - \dfrac{4\left(\frac{r+1}{r}\right)^4 K^2}{\gamma_{\epsilon x}\gamma_{\epsilon y}}}},$$
$$C = \sqrt{\frac{\gamma_{\epsilon x}\left(\rho^2 - \sqrt{\rho^4 - \dfrac{4\left(\frac{r+1}{r}\right)^4 K^2}{\gamma_{\epsilon x}\gamma_{\epsilon y}}}\right)}{\left(\frac{r+1}{r}\right)^2 K}}.$$

$\frac{\partial \zeta^B_t}{\partial \gamma_{\epsilon x}} < 0$ always holds when $r, K, \rho, \gamma_{\epsilon x}, \gamma_{\epsilon y}$ satisfy the equilibrium existence condition. □

*Proof of Lemma 3 (Comovement of $\Sigma_t^{-1}$ and $|\zeta_t|$).* The signal-to-noise ratio is $\zeta_t = \left[\bar{\Sigma}_t^{-1} -\right.$

$(r+1)\tau_y/r\big]\big/\big[(r+1)\zeta_t\tau_x/r-\rho\big]$, which leads to

$$\bar{\Sigma}_t^{-1}=\frac{r+1}{r}\tau_x\zeta_t^2-\rho\zeta_t+\frac{r+1}{r}\tau_y. \tag{C.2}$$

Taking the partial derivative with respect to $\zeta_t$, we have

$$\frac{\partial\bar{\Sigma}_t^{-1}}{\partial|\zeta_t|}=2\frac{r+1}{r}|\zeta_t|\tau_x+\rho>0.$$

The information processing capacity $K$, learning costs $\gamma_{\epsilon x}$, $\gamma_{\epsilon y}$ and $\gamma_{\nu x}$, and the prior precision of the dividend innovation $\tau_y$ affect $\bar{\Sigma}_t^{-1}$ solely through $|\zeta_t|$. Thus any change in $|\zeta_t|$ caused by $K$, $\gamma_{\epsilon x}$, $\gamma_{\epsilon y}$, $\gamma_{\nu x}$ or $\tau_y$ will be accompanied by a change in $\bar{\Sigma}_t^{-1}$ in the same direction. □

*Proof of Proposition 4.* Recall that $\bar{\Sigma}_t^{-1}=\frac{1}{r}\Sigma_t+\hat{\Sigma}_t$, which leads to $\bar{\Sigma}_t^{-1}=\frac{r+1}{r}\tau_y+\frac{1}{r}\tau_{\epsilon yt}+\hat{\tau}_{\epsilon yt}+\zeta_t^2\left(\frac{r+1}{r}\tau_x+\frac{1}{r}\tau_{\epsilon xt}+\hat{\tau}_{\epsilon xt}+\hat{\tau}_{\nu xt-1}\right)$. Plug it into (C.2), we have

$$\left(\frac{1}{r}\tau_{\epsilon xt}+\hat{\tau}_{\epsilon xt}+\hat{\tau}_{\nu xt-1}\right)\zeta_t^2+\rho\zeta_t+\frac{1}{r}\tau_{\epsilon yt}+\hat{\tau}_{\epsilon yt}=0. \tag{C.3}$$

This is a quadratic equation of $\zeta_t$, which has two solutions. We pick the larger solution, since when there is no demand information, the solution converges to the unique solution in the models where there is only dividend innovation acquisition.

$$\zeta_t=\frac{-\rho+\sqrt{\rho^2-4\left(\frac{1}{r}\tau_{\epsilon xt}+\hat{\tau}_{\epsilon xt}+\hat{\tau}_{\nu xt-1}\right)\left(\frac{1}{r}\tau_{\epsilon yt}+\hat{\tau}_{\epsilon yt}\right)}}{2\left(\frac{1}{r}\tau_{\epsilon xt}+\hat{\tau}_{\epsilon xt}+\hat{\tau}_{\nu xt-1}\right)}<0. \tag{C.4}$$

□

*Appendix C.1. Information Choices*

*Proof of Proposition 1.* (a) is obvious to see since $\hat{\tau}_{\nu xit}$ does not enter the *ex ante* utility.
(b) The Lagrangian is

$$\mathcal{L}=\hat{L}_{\epsilon y}\hat{\tau}_{\epsilon yit}+\hat{L}_{\epsilon y}\zeta_t^2\hat{\tau}_{\epsilon xit}-\lambda\left(\left[\gamma_{\epsilon x}\hat{\tau}_{\epsilon xit}^2+\gamma_{\epsilon y}\hat{\tau}_{\epsilon yit}^2\right]-K\right).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial \hat{\tau}_{\epsilon yit}} = \hat{L}_{\epsilon y} - 2\lambda \gamma_{\epsilon y} \hat{\tau}_{\epsilon yit} = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \hat{\tau}_{\epsilon xit}} = \hat{L}_{\epsilon y} \zeta_t^2 - 2\lambda \gamma_{\epsilon x} \hat{\tau}_{\epsilon xit} = 0,$$

which yields

$$\hat{\tau}_{\epsilon xit} = \frac{\gamma_{\epsilon y}}{\gamma_{\epsilon x}} \zeta_t^2 \hat{\tau}_{\epsilon yit}. \tag{C.5}$$

Plugging (C.5) into the budget constraint yields the optimal information choices. □

*Proof of Lemma 2.* To simplify the calculations, one can apply a strictly increasing transformation of the *ex ante* utility

$$V \overset{\text{def}}{=} -2 \ln \left[ \frac{-\mathbb{E}\left[ -\exp\left(-\rho W_{it+1^+} \mid \mathcal{I}_{t-1}^+\right)\right]}{\exp\left(-\rho r^2 W_0\right)} \right] = \ln H_1 + \ln H_2 + H_3 + H_4, \tag{C.6}$$

where

$$H_1 = \frac{\mathbb{V}ar\left(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}\right) + \Sigma_Z + \Sigma_a}{\mathbb{V}ar\left(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}\right)}, \tag{C.7}$$

$$H_2 = \frac{\mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right) + \Sigma_S}{\mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right)}, \tag{C.8}$$

$$H_3 = \frac{\bar{\bar{\Sigma}}_{t+1}^2 \rho^2 \bar{x}^2}{\mathbb{V}ar\left(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}\right) + \Sigma_Z + \Sigma_a}, \tag{C.9}$$

$$H_4 = \frac{\bar{\Sigma}_t^2 \rho^2 \bar{x}^2}{\mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right) + \Sigma_S}. \tag{C.10}$$

Notice that $\tau_{\epsilon yt}$ and $\tau_{\epsilon xt}$ enter only through $\mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right)$ and $\Sigma_S$ via $H_2$ and $H_4$. Thus we only need to check whether $\frac{\partial \mathbb{V}ar\left(\tilde{d}_t|\mathcal{I}_{it}\right)}{\partial \tau_{\epsilon yt}} = \frac{\partial \mathbb{V}ar\left(\tilde{d}_t|\mathcal{I}_{it}\right)}{\partial \tau_{\epsilon xt}} \zeta_t^2$ and $\frac{\partial \Sigma_S}{\partial \tau_{\epsilon yt}} = \frac{\partial \Sigma_S}{\partial \tau_{\epsilon xt}} \zeta_t^2$ hold. Recall that

$$\mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right) = \frac{1}{\tau_y + \tau_{\epsilon yt} + \zeta_t^2 \left(\tau_x + \tau_{\epsilon xt}\right)}.$$

Then

$$\frac{\partial \mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right)}{\partial \tau_{\epsilon yt}} = -\frac{1}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2}, \tag{C.11}$$

$$\frac{\partial \mathbb{V}ar\left(\tilde{d}_t \mid \mathcal{I}_{it}\right)}{\partial \tau_{\epsilon xt}} = -\frac{\zeta_t^2}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2}. \tag{C.12}$$

Thus $\frac{\partial \mathbb{V}ar\left(\tilde{d}_t|\mathcal{I}_{it}\right)}{\partial \tau_{\epsilon yt}} = \frac{\partial \mathbb{V}ar\left(\tilde{d}_t|\mathcal{I}_{it}\right)}{\partial \tau_{\epsilon xt}}\zeta_t^2$ holds. One has

$$\Sigma_S = \frac{b_2^2}{\tau_y} + \frac{b_3^2}{\tau_x} + \frac{\tau_{\epsilon yt} + \zeta_t^2\tau_{\epsilon xt}}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} \tag{C.13}$$

Taking the partial derivative with respect to $\tau_{\epsilon yt}$ on both sides, we have

$$\begin{aligned}\frac{\partial \Sigma_S}{\partial \tau_{\epsilon yt}} = &\frac{2b_2}{\tau_y}\frac{\tau_y}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} - \frac{2b_3}{\tau_x}\frac{\zeta_t\tau_x}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} \\ &+ \frac{1}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} - 2\frac{\tau_{\epsilon yt} + \zeta_t^2\tau_{\epsilon xt}}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^3},\end{aligned} \tag{C.14}$$

$$\begin{aligned}\frac{\partial \Sigma_S}{\partial \tau_{\epsilon xt}} = &\frac{2b_2}{\tau_y}\frac{\zeta_t^2\tau_y}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} - \frac{2b_3}{\tau_x}\frac{\zeta_t^3\tau_x}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} \\ &+ \frac{\zeta_t^2}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^2} - 2\zeta_t^2\frac{\tau_{\epsilon yt} + \zeta_t^2\tau_{\epsilon xt}}{\left[\tau_y + \tau_{\epsilon yt} + \zeta_t^2\left(\tau_x + \tau_{\epsilon xt}\right)\right]^3}.\end{aligned} \tag{C.15}$$

Therefore $\frac{\partial \Sigma_S}{\partial \tau_{\epsilon yt}} = \frac{\partial \Sigma_S}{\partial \tau_{\epsilon xt}}\zeta_t^2$ holds. □

*Proof of Proposition 2.*

Step 1: Calculate learning indices

On the one hand, from the proof of Lemma 2, we know that the learning index associated with the signal for the current fundamental shock is

$$L_{\epsilon yit} = \partial \ln H_2/\partial \tau_{\epsilon yt} + \partial H_4/\partial \tau_{\epsilon yt},$$

with

$$\frac{\partial \ln H_2}{\partial \tau_{\epsilon yt}} = \frac{1}{\left[\mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it}) + \Sigma_S\right]\mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it})} \times \left[\frac{\partial \Sigma_S}{\partial \tau_{\epsilon yt}}\mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it}) - \frac{\partial \mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it})}{\partial \tau_{\epsilon yt}}\Sigma_S\right],$$

$$\frac{\partial H_4}{\partial \tau_{\epsilon yt}} = -\frac{\bar{\Sigma}_t^2 \rho^2 \bar{x}^2}{\left[\mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it}) + \Sigma_S\right]^2}\left[\frac{\partial \Sigma_S}{\partial \tau_{\epsilon yt}} + \frac{\partial \mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it})}{\partial \tau_{\epsilon yt}}\right].$$

where $\partial \Sigma_S / \partial \tau_{\epsilon yt}$ and $\partial \mathbb{V}ar(\tilde{d}_t \mid \mathcal{I}_{it}) / \partial \tau_{\epsilon yt}$ are calculated by (C.14) and (C.11).

On the other hand, notice that $\tau_{\nu xt}$ enter the ex-ante utility function only through $\mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1})$, $\Sigma_Z$ and $\Sigma_a$ via $H_1$ and $H_3$. Thus

$$L_{\nu xit} = \partial \ln H_1 / \partial \tau_{\nu xt} + \partial H_3 / \partial \tau_{\nu xt},$$

with

$$\frac{\partial \ln H_1}{\partial \tau_{\nu xt}} = \frac{\frac{\partial (\Sigma_Z + \Sigma_a)}{\partial \tau_{\nu xt}}\mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}) - \frac{\partial \mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1})}{\partial \tau_{\nu xt}}(\Sigma_Z + \Sigma_a)}{\left[\mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}) + \Sigma_Z + \Sigma_a\right]\mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1})},$$

$$\frac{\partial H_3}{\partial \tau_{\nu xt}} = -\frac{\bar{\bar{\Sigma}}_{t+1}^2 \rho^2 \bar{x}^2}{\left[\mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}) + \Sigma_Z + \Sigma_a\right]^2} \times \left[\frac{\partial (\Sigma_Z + \Sigma_a)}{\partial \tau_{\nu xt}} + \frac{\partial \mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1})}{\partial \tau_{\nu xt}}\right].$$

Recall that

$$\mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1}) = 1 \Big/ \big(\tau_y + \hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2(\tau_x + \hat{\tau}_{\epsilon xit+1} + \tau_{\nu xt})\big).$$

Thus

$$\frac{\partial \mathbb{V}ar(\tilde{d}_{t+1} \mid \hat{\mathcal{I}}_{it+1})}{\partial \tau_{\nu xt}} = -\zeta_{t+1}^2 \Big/ \left[\tau_y + \hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2(\tau_x + \hat{\tau}_{\epsilon xit+1} + \tau_{\nu xt})\right]^2.$$

One has

$$\Sigma_Z + \Sigma_a = \frac{a_2^2}{\tau_y} + \frac{a_3^2 + c^2}{\tau_x + \tau_{\nu xt}} + \frac{\hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2 \hat{\tau}_{\epsilon xit+1}}{\left[\tau_y + \hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2(\tau_x + \hat{\tau}_{\epsilon xit+1} + \tau_{\nu xt})\right]^2}.$$

Taking the partial derivative with respect to $\tau_{\nu xt}$ on both sides, we have

$$\begin{aligned}\frac{\partial(\Sigma_Z + \Sigma_a)}{\partial \tau_{\nu xt}} = \frac{2a_2}{\tau_y} \frac{\zeta_{t+1}^2 \tau_y}{\left[\tau_y + \hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2(\tau_x + \hat{\tau}_{\epsilon xit+1} + \tau_{\nu xt})\right]^2} - \frac{a_3^2 + c^2}{(\tau_x + \tau_{\nu xt})^2} \\ + \frac{2a_3}{\tau_x + \tau_{\nu xt}} \left( \frac{\zeta_{t+1}}{D} - \frac{\zeta_{t+1}^3(\tau_x + \tau_{\nu xt})}{D^2} \right) \\ + \frac{2c}{\tau_x + \tau_{\nu xt}} \left( \frac{\zeta_{t+1}}{D} - \frac{\zeta_{t+1}^3 \tau_x}{D^2} \right) - 2\zeta_{t+1}^2 \frac{\hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2 \hat{\tau}_{\epsilon xit+1}}{D^3},\end{aligned}$$

where $D \triangleq \tau_y + \hat{\tau}_{\epsilon yit+1} + \zeta_{t+1}^2(\tau_x + \hat{\tau}_{\epsilon xit+1} + \tau_{\nu xt})$ for notational brevity.

Step 2: Characterizing information choices with F.O.C.'s and information capacity constraint

The first-order conditions for the information choice problem are

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \tau_{\epsilon yt}} &= L_{\epsilon yit} - 2\lambda \gamma_{\epsilon y} \tau_{\epsilon yt} = 0, \\ \frac{\partial \mathcal{L}}{\partial \tau_{\epsilon xt}} &= L_{\epsilon yit} \zeta_t^2 - 2\lambda \gamma_{\epsilon x} \tau_{\epsilon xt} = 0, \\ \frac{\partial \mathcal{L}}{\partial \tau_{\nu xt}} &= L_{\nu xit} - 2\lambda \gamma_{\nu x} \tau_{\nu xt} = 0.\end{aligned}$$

Dividing the first-order conditions by one another, we have

$$\begin{aligned}\tau_{\epsilon yt} &= \frac{\gamma_{\nu x}}{\gamma_{\epsilon y}} \frac{L_{\epsilon yit}}{L_{\nu xit}} \tau_{\nu xt}, \\ \tau_{\epsilon xt} &= \frac{\gamma_{\nu x}}{\gamma_{\epsilon x}} \frac{L_{\epsilon yit}}{L_{\nu xit}} \zeta_t^2 \tau_{\nu xt}.\end{aligned}$$

Substituting these expressions into the budget constraint (4), we have

$$\frac{\tau_{\nu xt}^2}{L_{\nu xit}^2} \left( \frac{\gamma_{\nu x}}{\gamma_{\epsilon y}} L_{\epsilon yit}^2 + \frac{\gamma_{\nu x}}{\gamma_{\epsilon x}} L_{\epsilon yit}^2 \zeta_t^4 + L_{\nu xit}^2 \right) = \frac{K}{\gamma_{\nu x}}. \tag{C.16}$$

Define

$$f_{it} \equiv \frac{L_{\epsilon yit}}{L_{\nu xit}} = \frac{\partial \ln H_2/\partial \tau_{\epsilon yt} + \partial H_4/\partial \tau_{\epsilon yt}}{\partial \ln H_1/\partial \tau_{\nu xt} + \partial H_3/\partial \tau_{\nu xt}}. \tag{C.17}$$

Then Eq.(C.16) simplifies to

$$\tau_{\nu xt}^2 \left( \frac{\gamma_{\nu x}}{\gamma_{\epsilon y}} f_{it}^2 + \frac{\gamma_{\nu x}}{\gamma_{\epsilon x}} \zeta_t^4 f_{it}^2 + 1 \right) = \frac{K}{\gamma_{\nu x}}.$$

□

*Proof of Theorem 3 (Price Volatility).* The unconditional price volatility is $\mathbb{V}ar(p_t) = C_t^2 \tau_y^{-1} + D_t^2 \tau_x^{-1}$. Taking the partial derivative with respect to $K$, we have

$$\frac{\partial \mathbb{V}ar(p_t)}{\partial K} = 2 \left[ \frac{C_t}{\tau_y} \frac{\partial C_t}{\partial \bar{\Sigma}_t} \frac{\partial \bar{\Sigma}_t}{\partial \zeta_t} + \frac{D_t}{\tau_x} \left( \frac{\partial D_t}{\partial \bar{\Sigma}_t} \frac{\partial \bar{\Sigma}_t}{\partial \zeta_t} + \frac{\partial D_t}{\partial \zeta_t} \right) \right] \frac{\partial \zeta_t}{\partial K}. \tag{C.18}$$

We then need to determine the sign of the term inside the square brackets. To do that, we first derive the partial derivatives of $C_t$ and $D_t$ with respect to $\bar{\Sigma}_t$ and $\zeta_t$. We have $\frac{\partial C_t}{\partial \bar{\Sigma}_t} = -\frac{r+1}{r}\tau_y$, $\frac{\partial D_t}{\partial \bar{\Sigma}_t} = \frac{r+1}{r}\zeta_t \tau_x - \rho$, $\frac{\partial D_t}{\partial \zeta_t} = \frac{r+1}{r}\bar{\Sigma}_t \tau_x$ and $\frac{\partial \bar{\Sigma}_t}{\partial \zeta_t} = \bar{\Sigma}_t^2 \left(\rho - 2\frac{r+1}{r}\zeta_t \tau_x\right)$. Thus the term inside square brackets becomes

$$\begin{aligned}
&\frac{C_t}{\tau_y} \frac{\partial C_t}{\partial \bar{\Sigma}_t} \frac{\partial \bar{\Sigma}_t}{\partial \zeta_t} + \frac{D_t}{\tau_x} \Big( \frac{\partial D_t}{\partial \bar{\Sigma}_t} \frac{\partial \bar{\Sigma}_t}{\partial \zeta} + \frac{\partial D_t}{\partial \zeta_t} \Big) \\
=&\frac{1 - \frac{r+1}{r}\bar{\Sigma}_t \tau_y}{\tau_y} \Big( -\frac{r+1}{r}\tau_y \Big) \bar{\Sigma}_t^2 \Big( \rho - 2\frac{r+1}{r}\zeta_t \tau_x \Big) \\
&+ \frac{\frac{r+1}{r}\bar{\Sigma}_t \zeta_t \tau_x - \rho \bar{\Sigma}_t}{\tau_x} \left[ \Big( \frac{r+1}{r}\zeta_t \tau_x - \rho \Big) \bar{\Sigma}_t^2 \Big( \rho - 2\frac{r+1}{r}\zeta_t \tau_x \Big) + \frac{r+1}{r}\bar{\Sigma}_t \tau_x \right] \\
=&\bar{\Sigma}_t^2 \Bigg\{ \frac{r+1}{r} \Big( 1 - \frac{r+1}{r}\bar{\Sigma}_t \tau_y \Big) \Big( 2\frac{r+1}{r}\zeta_t \tau_x - \rho \Big) \\
&+ \Big( \frac{r+1}{r}\zeta_t \tau_x - \rho \Big) \left[ \Big( \frac{r+1}{r}\zeta_t \tau_x - \rho \Big) \frac{\bar{\Sigma}_t}{\tau_x} \Big( \rho - 2\frac{r+1}{r}\zeta_t \tau_x \Big) + \frac{r+1}{r} \right] \Bigg\}.
\end{aligned}$$

Let

$$\begin{aligned}\mathcal{A} \equiv & \Big[\frac{r+1}{r}\Big]\Big(1-\Big[\frac{r+1}{r}\Big]\bar{\Sigma}_t\tau_y\Big)\Big(2\Big[\frac{r+1}{r}\Big]\zeta_t\tau_x-\rho\Big) \\ & +\Big(\Big[\frac{r+1}{r}\Big]\zeta_t\tau_x-\rho\Big)\times \\ & \quad\Bigg[\Big(\Big[\frac{r+1}{r}\Big]\zeta_t\tau_x-\rho\Big)\big(\bar{\Sigma}_t/\tau_x\big)\Big(\rho-2\Big[\frac{r+1}{r}\Big]\zeta_t\tau_x\Big)+\frac{r+1}{r}\Bigg].\end{aligned}$$

For a given parameter region, we determine the sign of $\mathcal{A}$.

For $\mathcal{A} \leq 0$, $\zeta < 0$, $\rho, \tau_x, \tau_y > 0$ and $\bar{\Sigma}_t^{-1} = \frac{r+1}{r}\tau_y + \frac{r+1}{r}\tau_x\zeta_t^2 - \rho\zeta_t > 0$ to hold simultaneously, we must have

$$\begin{aligned}0 < \bar{\Sigma}_t \leq & \frac{-2\,\dfrac{r+1}{r}\tau_x\rho + 3\left(\dfrac{r+1}{r}\right)^2\tau_x^2\zeta_t}{-\left(\dfrac{r+1}{r}\right)^2\tau_x\tau_y\rho - \rho^3 + 2\left(\dfrac{r+1}{r}\right)^3\tau_x^2\tau_y\zeta_t + 4\left(\dfrac{r+1}{r}\right)\tau_x\rho^2\zeta_t} \\ & - 5\left(\frac{r+1}{r}\right)^2\tau_x^2\rho\zeta_t^2 + 2\left(\frac{r+1}{r}\right)^3\tau_x^3\zeta_t^3.\end{aligned}$$

Substituting the expression of $\bar{\Sigma}_t$ and solving for $\zeta_t$, we obtain the feasible sets:

- $\{\zeta_t \mid \zeta_t < 0\}$, if $\tau_x \geq \bar{\Omega}_x$;
- $\{\zeta_t \mid \zeta_t \leq \underline{\zeta}_t \text{ or } \bar{\zeta}_t \leq \zeta_t < 0\}$, if $\underline{\Omega}_x < \tau_x < \bar{\Omega}_x$;
- $\{\zeta_t \mid \zeta_t \leq \underline{\zeta}_t\}$, if $0 < \tau_x \leq \underline{\Omega}_x$.

Since $\partial\zeta_t/\partial K < 0$, the sign of $\partial\mathbb{V}ar(p_t)/\partial K$ is:

(a) if $\tau_x \geq \bar{\Omega}_x$, then $\partial\mathbb{V}ar(p_t)/\partial K \geq 0$;

(b) if $\underline{\Omega}_x < \tau_x < \bar{\Omega}_x$, then

$$\frac{\partial\mathbb{V}ar(p_t)}{\partial K} = \begin{cases} \geq 0, & \text{if } \zeta_t \leq \underline{\zeta}_t \text{ or } \bar{\zeta}_t \leq \zeta_t < 0, \\ < 0, & \text{if } \underline{\zeta}_t < \zeta_t < \bar{\zeta}_t; \end{cases}$$

(c) If $0 < \tau_x \leq \underline{\Omega}_x$, then

$$\frac{\partial \mathbb{V}ar(p_t)}{\partial K} = \begin{cases} \geq 0, & \text{if } \zeta_t \leq \underline{\zeta}_t, \\ < 0, & \text{if } \underline{\zeta}_t < \zeta_t < 0; \end{cases}$$

where

$$\begin{aligned} \bar{\Omega}_x &\equiv 5\rho^2 / \left(4 \left(\frac{r+1}{r}\right)^2 \tau_y\right), \\ \underline{\Omega}_x &\equiv \rho^2 / \left( \left(\frac{r+1}{r}\right)^2 \tau_y\right), \\ \bar{\zeta}_t &\equiv \frac{r}{2(r+1)\tau_x} \left[ -\rho + \sqrt{-4 \left(\frac{r+1}{r}\right)^2 \tau_x \tau_y + 5\rho^2} \right], \\ \underline{\zeta}_t &\equiv \frac{r}{2(r+1)\tau_x} \left[ -\rho - \sqrt{-4 \left(\frac{r+1}{r}\right)^2 \tau_x \tau_y + 5\rho^2} \right]. \end{aligned}$$

□

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